Masalah Penugasan (*Assignment Problem*)

Kuliah 14

Materi Bahasan

① Rumusan masalah penugasan

② Pemecahan masalah penugasan

① Rumusan Masalah Penugasan

Masalah Penugasan

- Terdapat n mesin (M_1, M_2, …, M_n).
- Terdapat n job (J_1, J_2, …, J_n) yang harus ditugaskan (dikerjakan) pada mesin.
- Biaya pengerjaan untuk setiap job tergantung pada mesin yang ditugaskan.
- Biaya pengerjaan job j pada mesin i dinyatakan dengan c_{ij}. Jika suatu job i tidak dapat ditugaskan pada mesin j, maka $c_{ij} = M$ (M bilangan positif yang sangat besar)
- Tiap mesin hanya dapat mengerjakan satu job.
- Permasalahannya adalah menentukan penugasan job pada mesin yang meminimumkan biaya total.

Rumusan Pemrograman Linier Bilangan Bulat Biner

Meminimumkan $Z = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$

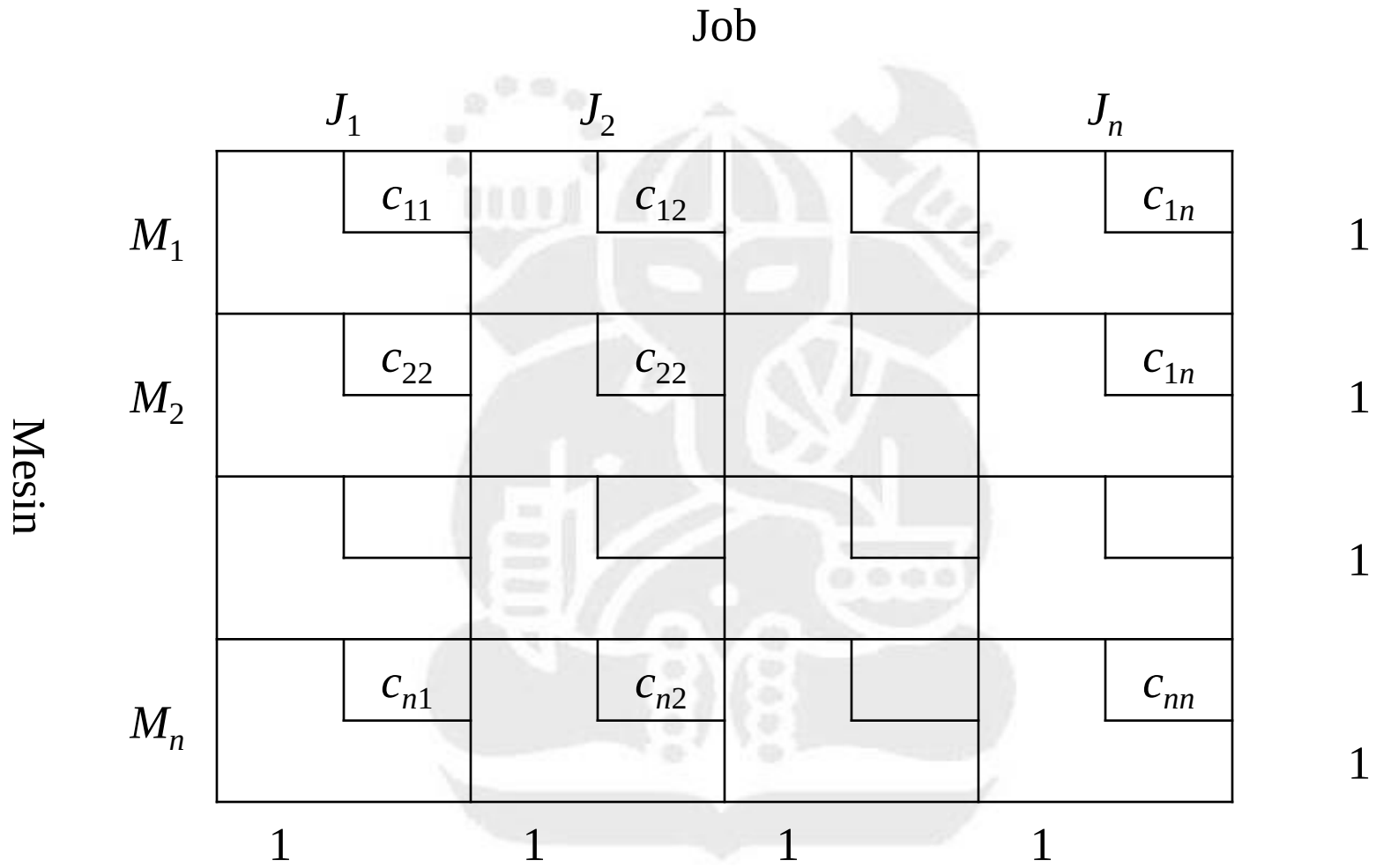
dengan pembatas-pembatas:

$$\sum_{j=1}^{n} x_{ij} = 1, \qquad i = 1, 2, \ldots, n$$

$$\sum_{i=1}^{n} x_{ij} = 1, \qquad j = 1, 2, \ldots, n$$

$$x_{ij} = \begin{cases} 1 & \text{jika job } j \text{ ditugaskan pada mesin } i \\ 0 & \text{yang lain} \end{cases}$$

Tabel Transportasi



Masalah Penugasan Tak Seimbang

- Jika jumlah mesin lebih banyak dari jumlah job ($M > N$) maka dibuat tambahan ($M - N$) job fiktif dengan biaya penugasan nol.
- Jika jumlah mesin lebih sedikit dari jumlah job ($M < N$) maka dibuat tambahan ($N - M$) mesin fiktif dengan biaya penugasan nol.

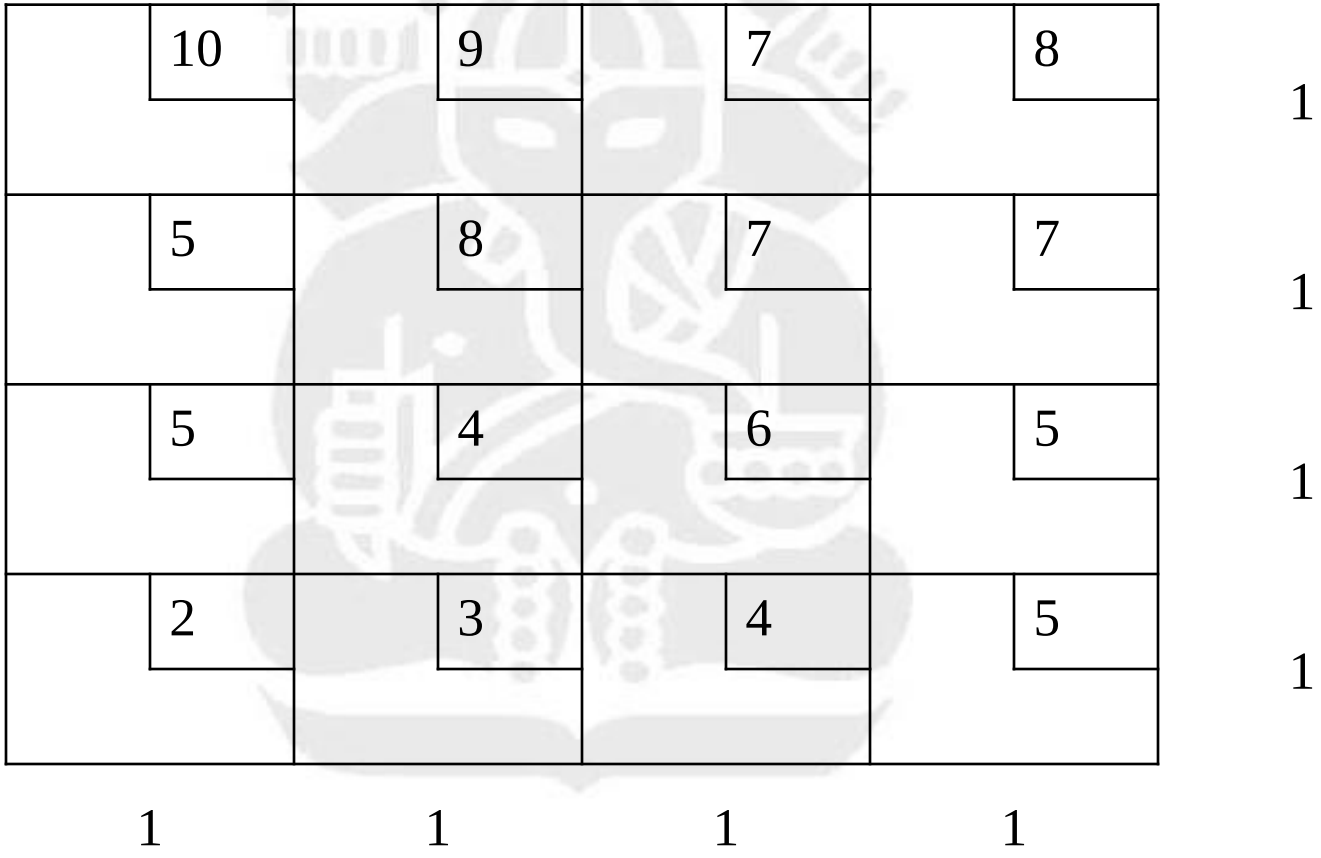
② Pemecahan Masalah Penugasan

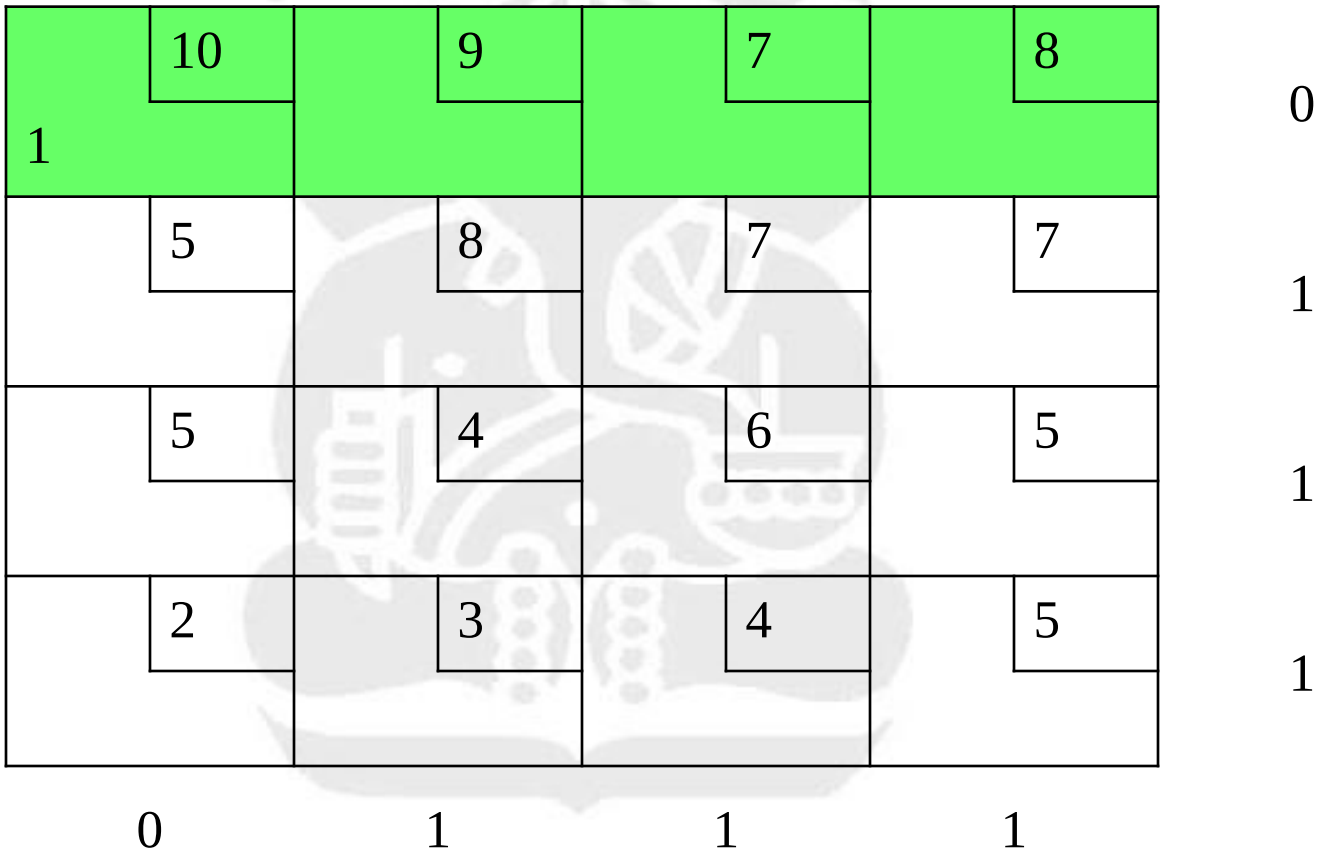
Teknik Pemecahan Masalah Penugasan

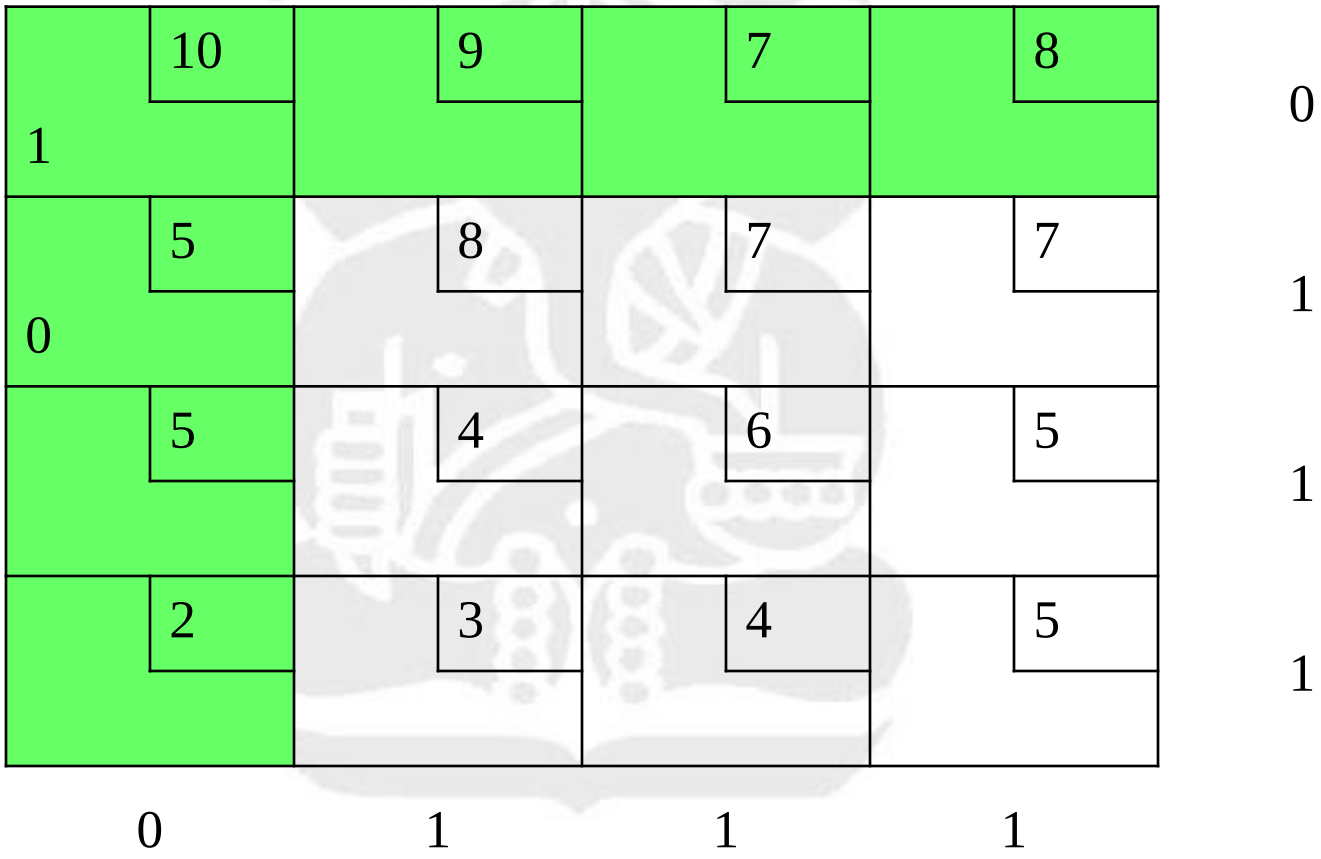
- Teknik untuk pemecahan masalah transportasi
  - Kendala: adanya degenerasi
- *Branch-and-bound*
- Pemrograman dinamis (Penelitian Operasional II)
- Metode Hungarian (*Hungarian Method*)

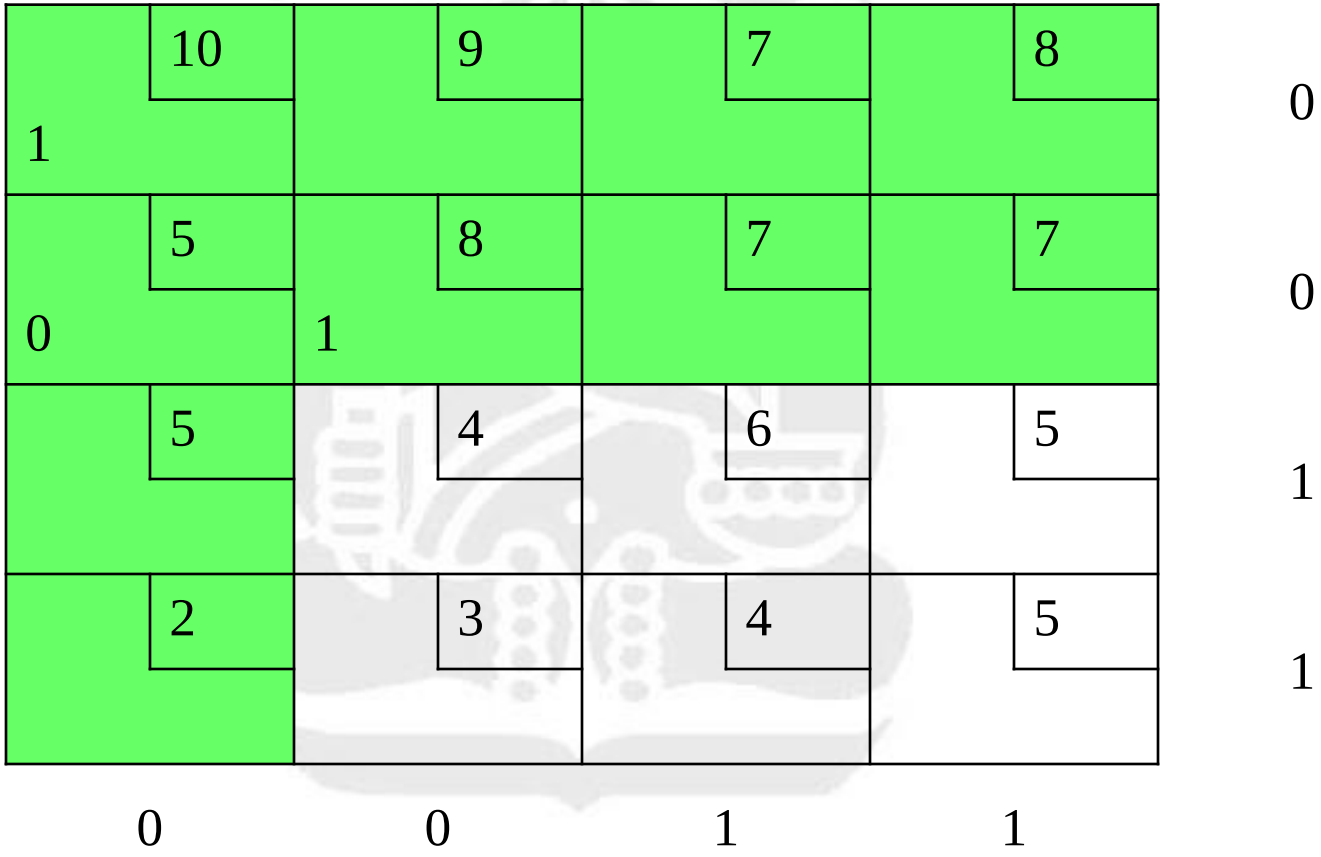
Pemecahan dengan Teknik Pemecahan untuk Masalah Transportasi

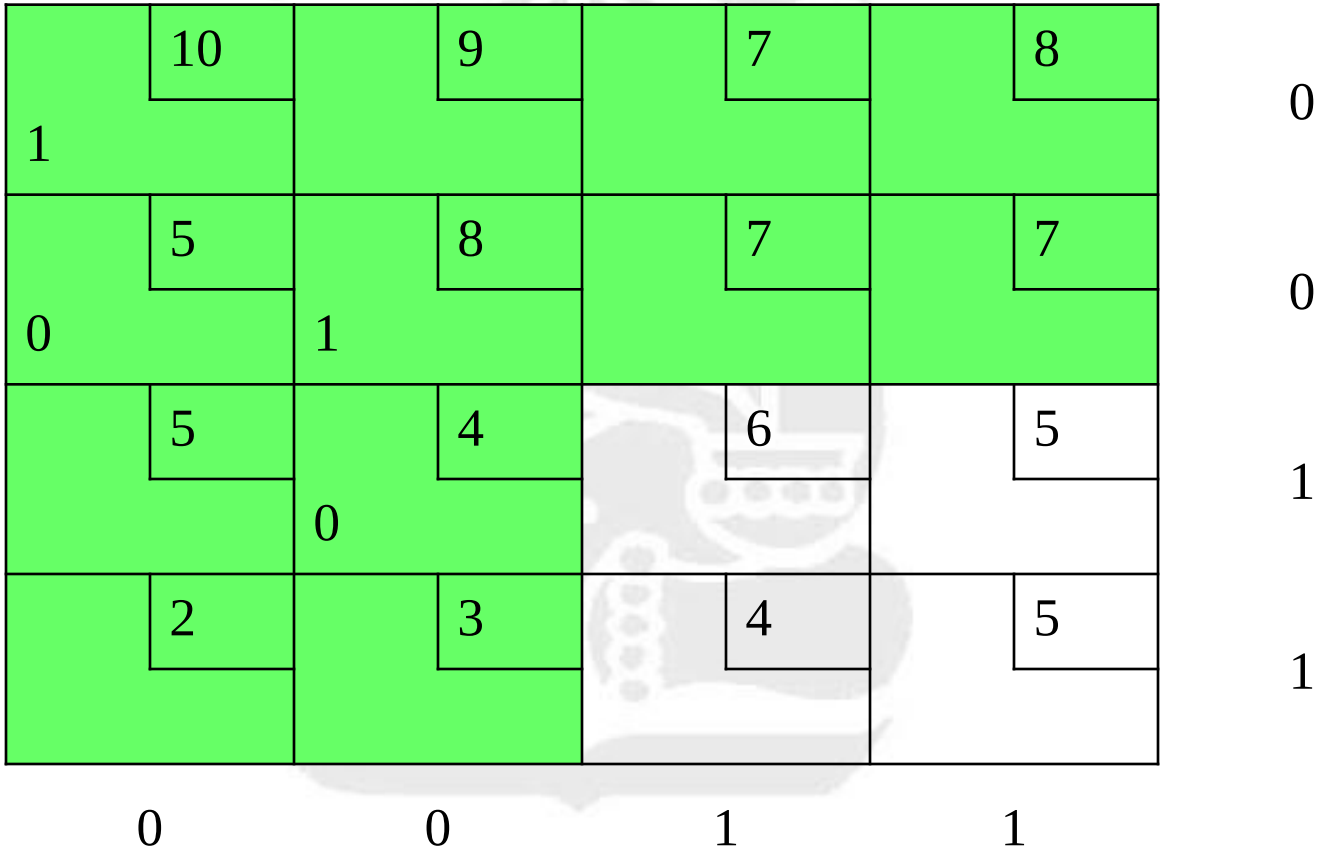
	J_1	J_2	J_3	J_4	
M_1	10	9	7	8	1
M_2	5	8	7	7	1
M_3	5	4	6	5	1
M_4	2	3	4	5	1
	1	1	1	1	

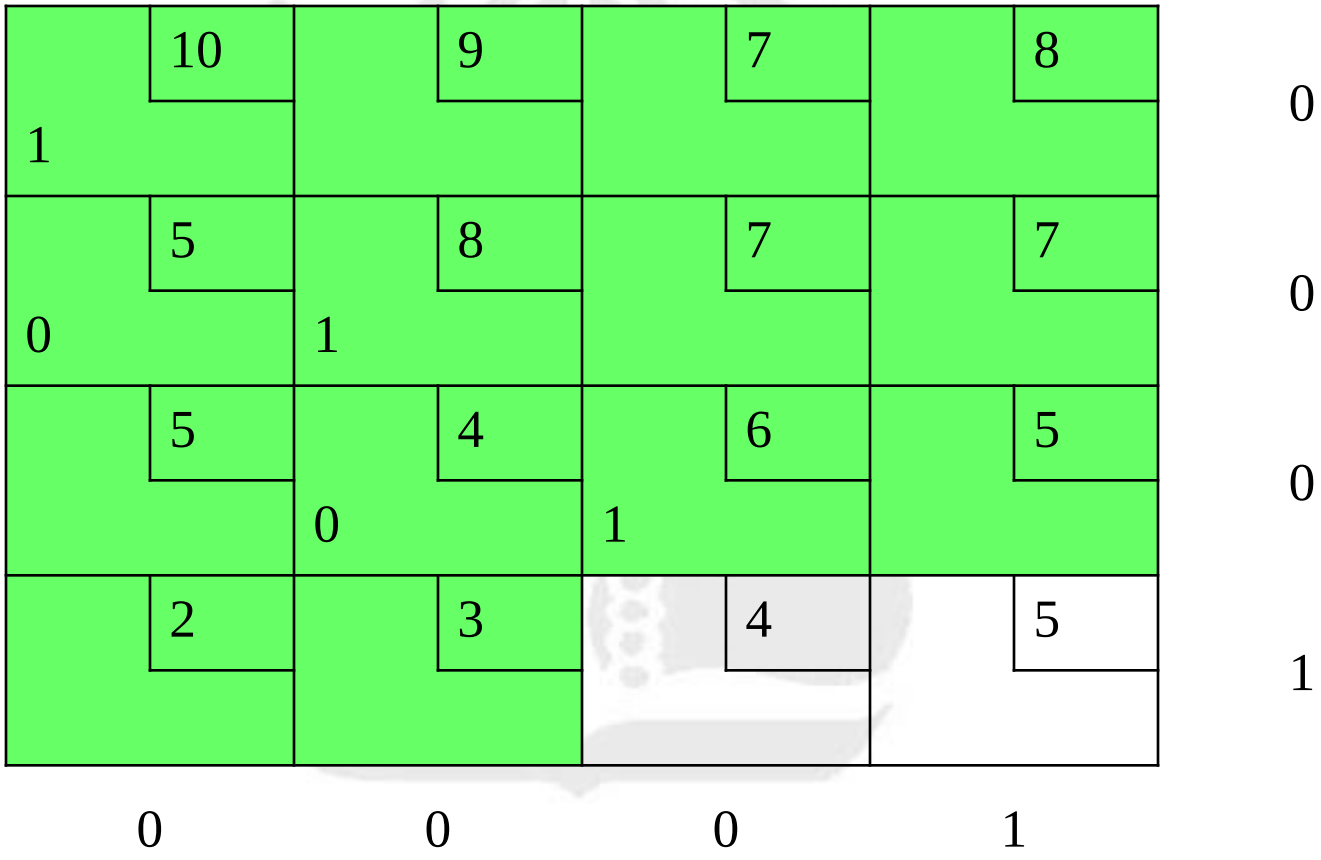

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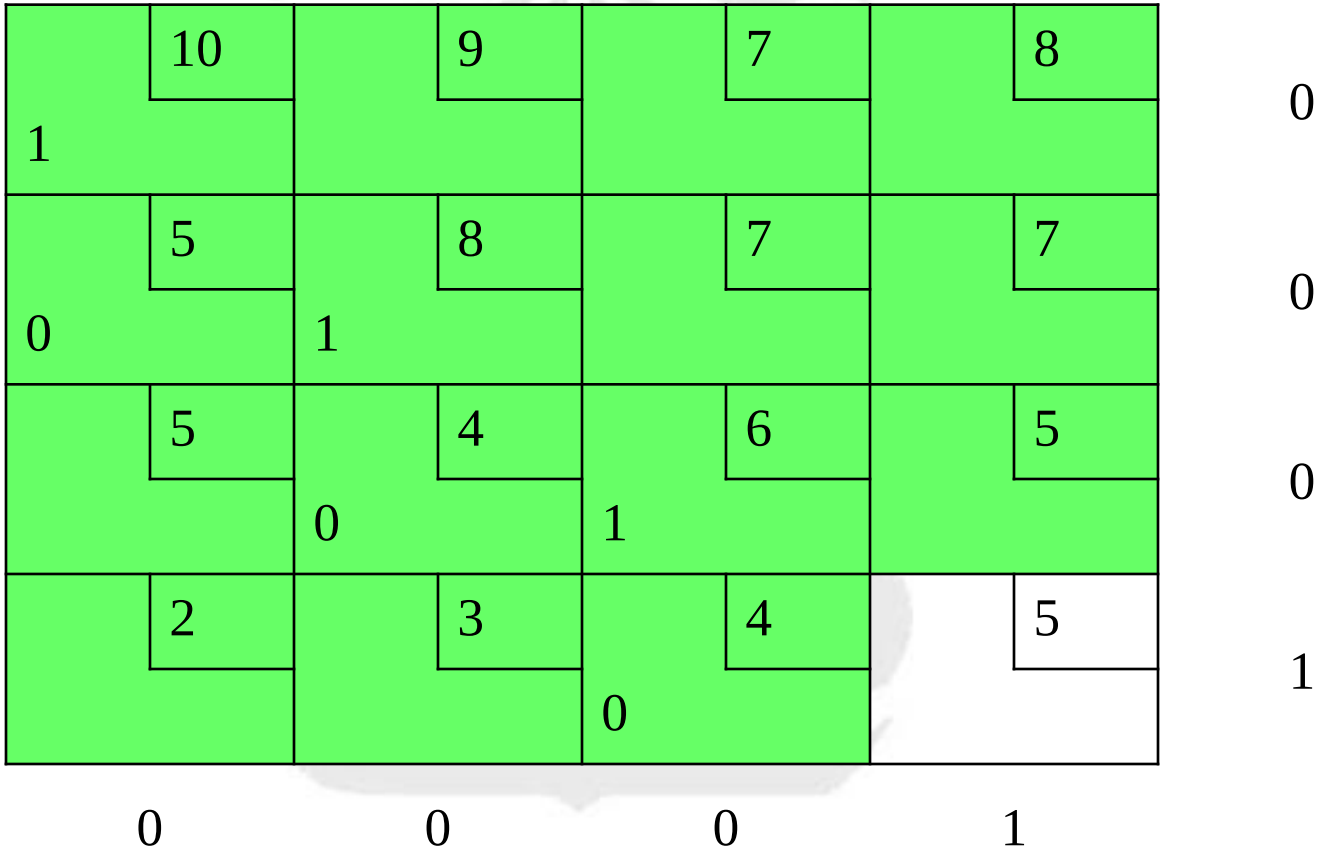

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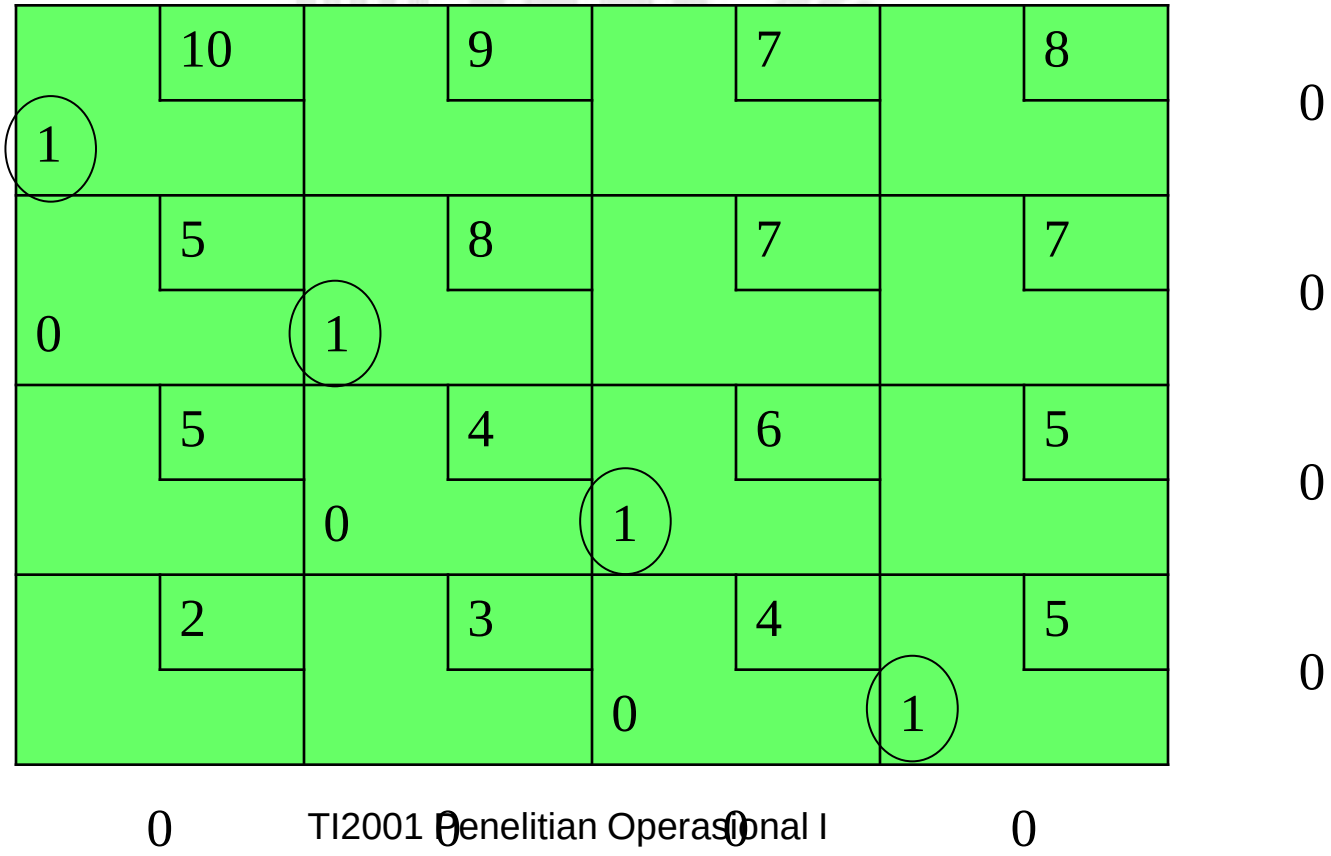

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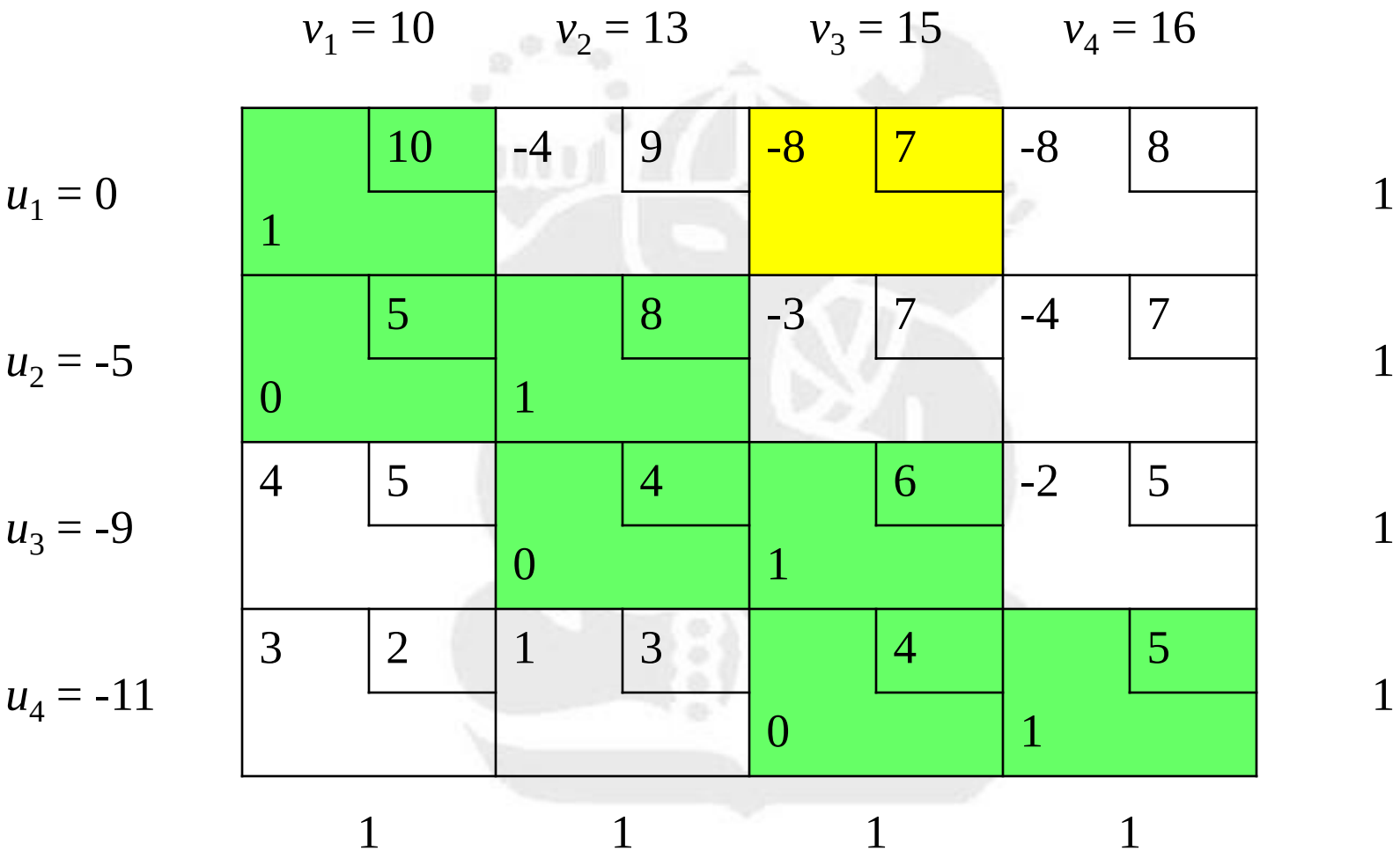

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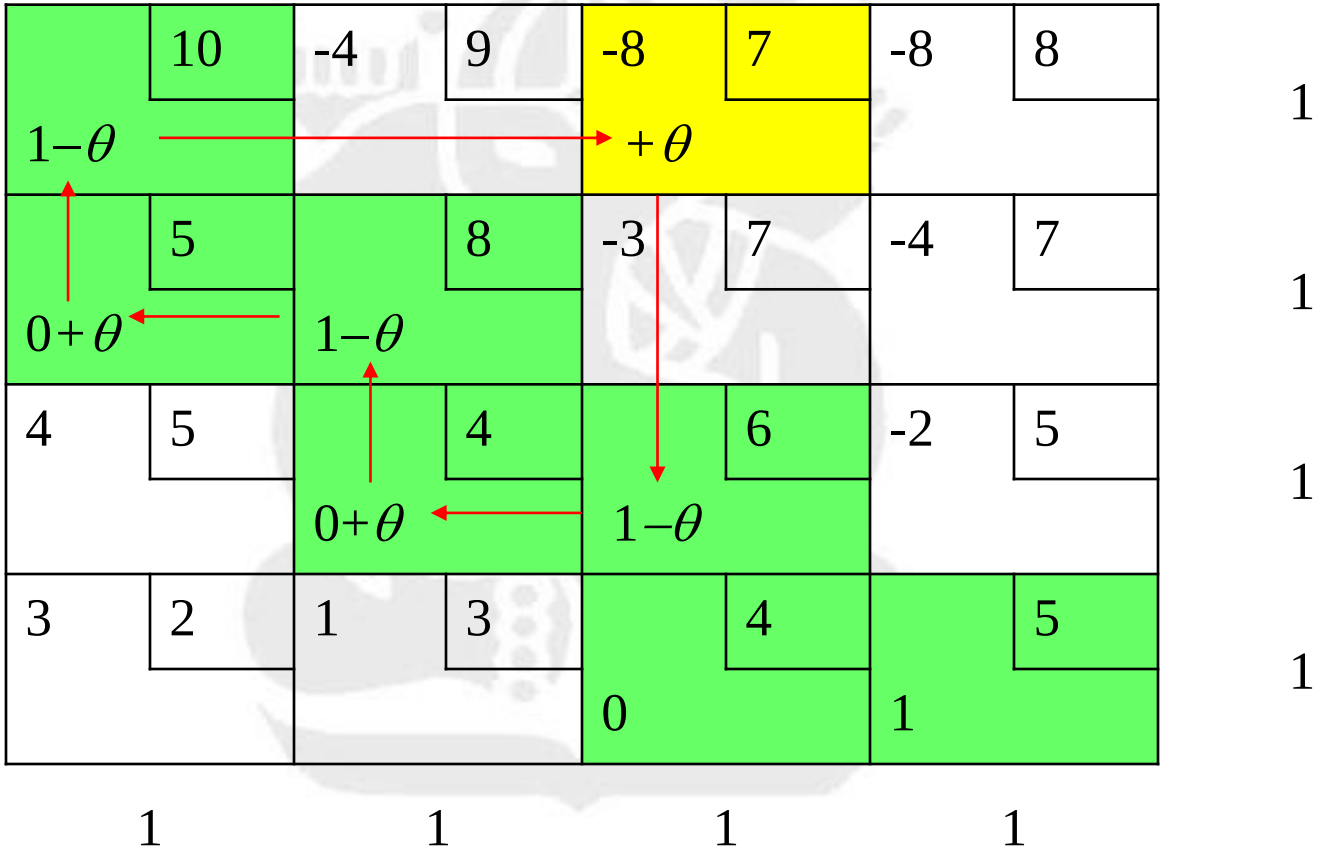
Solusi basis layak awal

$M_1 \rightarrow J_1$
$M_2 \rightarrow J_2$
$M_3 \rightarrow J_3$
$M_4 \rightarrow J_4$

$$Z = 10(1) + 8(1) + 6(1) + 5(1)$$
$$= 29$$



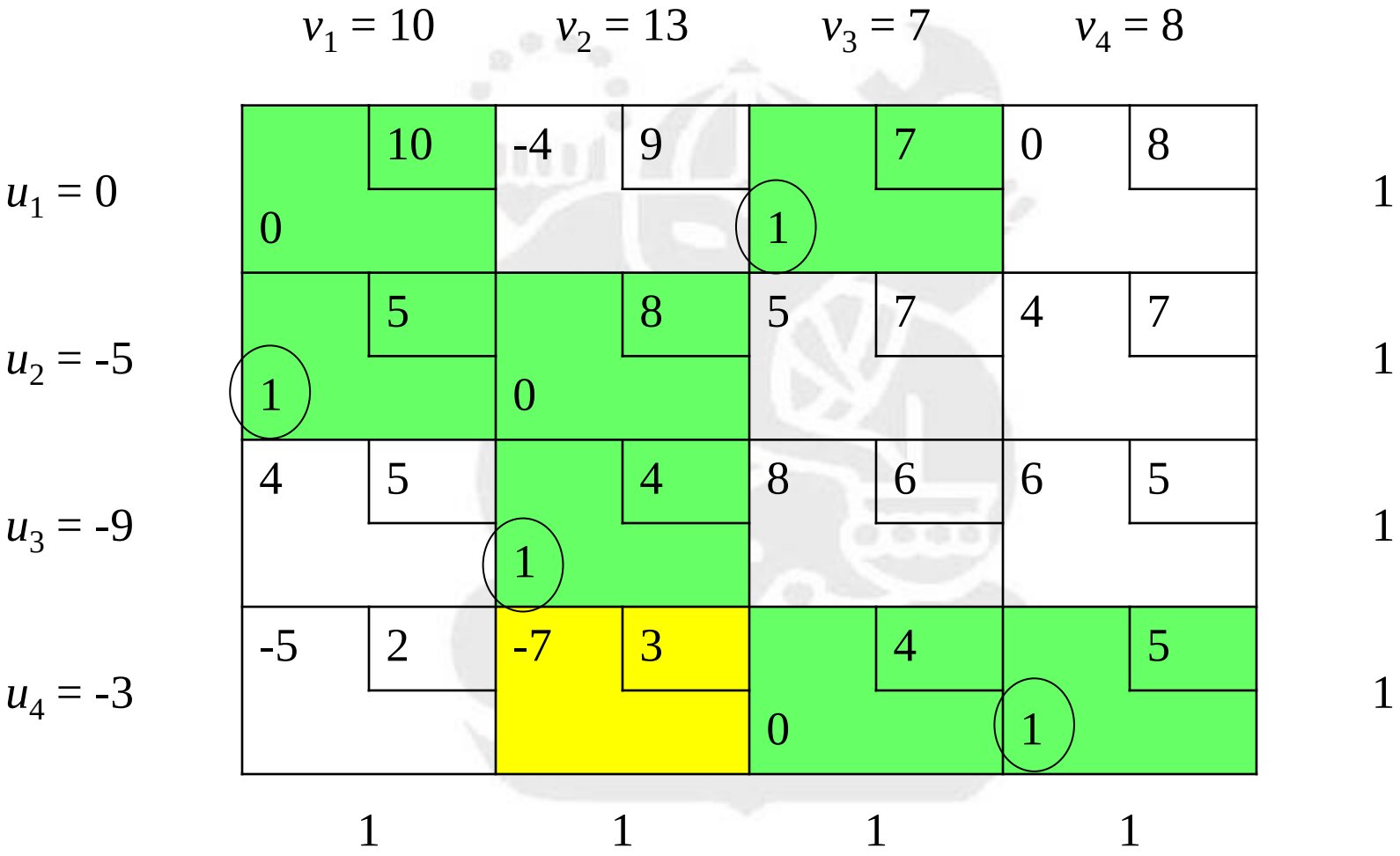

$v_1 = 10$
$v_2 = 13$
$v_3 = 15$
$v_4 = 16$
$u_1 = 0$
$u_2 = -5$
$u_3 = -9$
$u_4 = -11$
10
-4
9
-8
7
-8
8
1
1
5
8
-3
7
-4
7
0
1
1
4
5
4
6
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0
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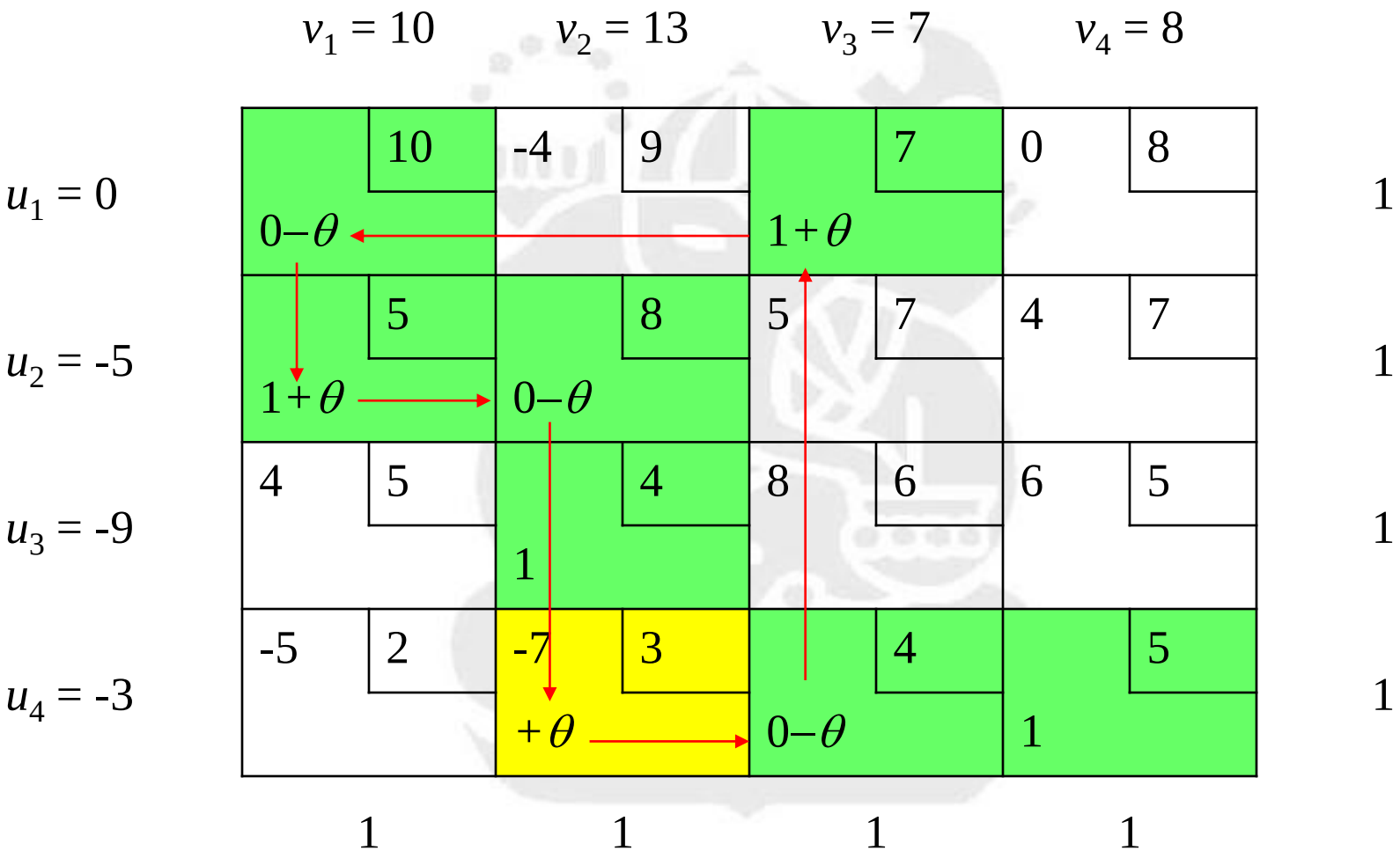

10
-4
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-8
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1−θ
+θ
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-3
7
-4
7
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1−θ
4
5
4
6
-2
5
0+θ
1−θ
3
2
1
3
4
5
0
1
1
1
1
1
1
1
1

Solusi basis layak baru

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_4$

$Z = 7(1) + 5(1) + 4(1) + 5(1)$
$= 21$

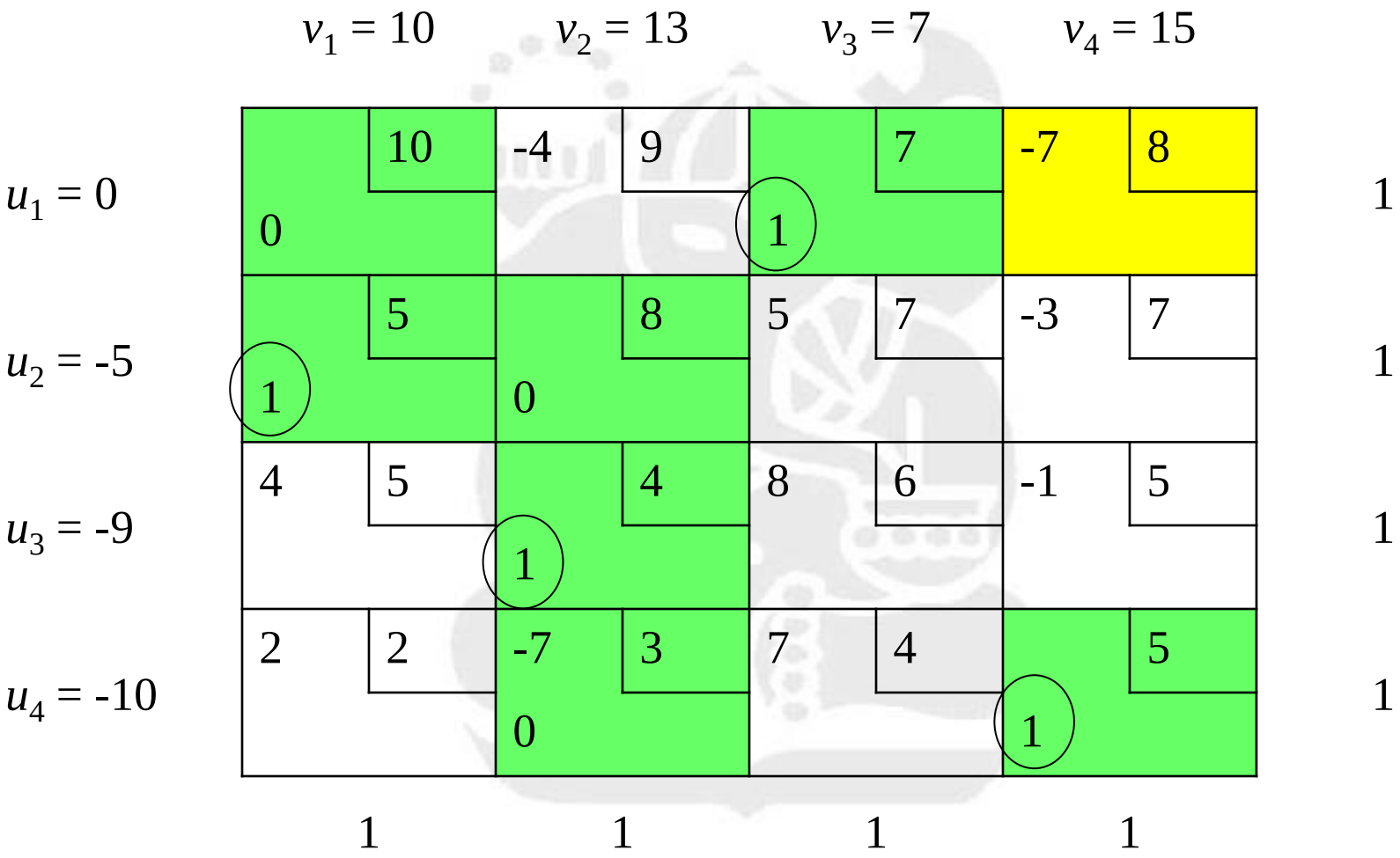


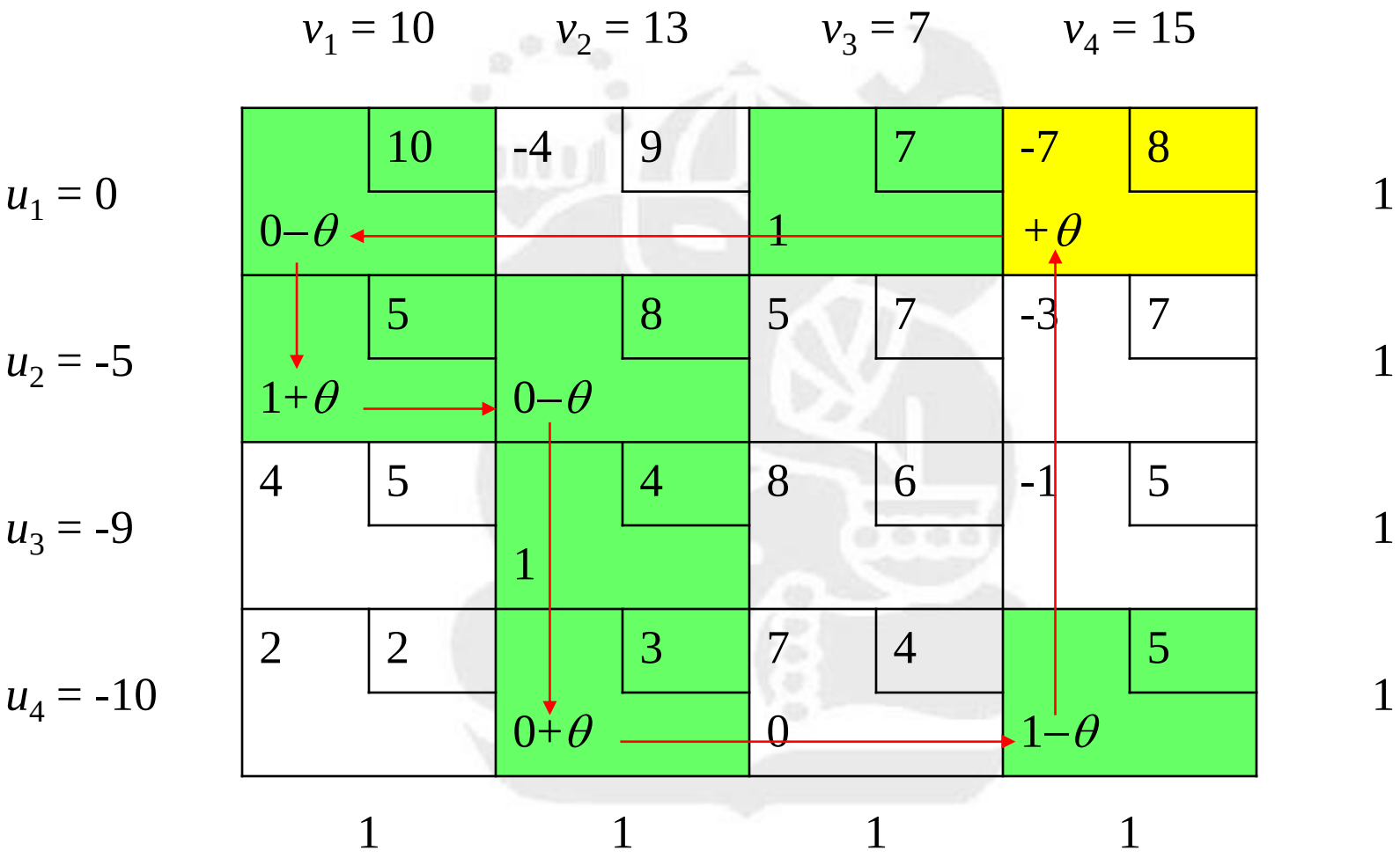

$v_1 = 10$
$v_2 = 13$
$v_3 = 7$
$v_4 = 8$
$u_1 = 0$
$u_2 = -5$
$u_3 = -9$
$u_4 = -3$
10
-4
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$1+\theta$
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$0-\theta$
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-7
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$0-\theta$
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Solusi basis layak baru

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_4$

$Z = 7(1) + 5(1) + 4(1) + 5(1)$
$= 21$

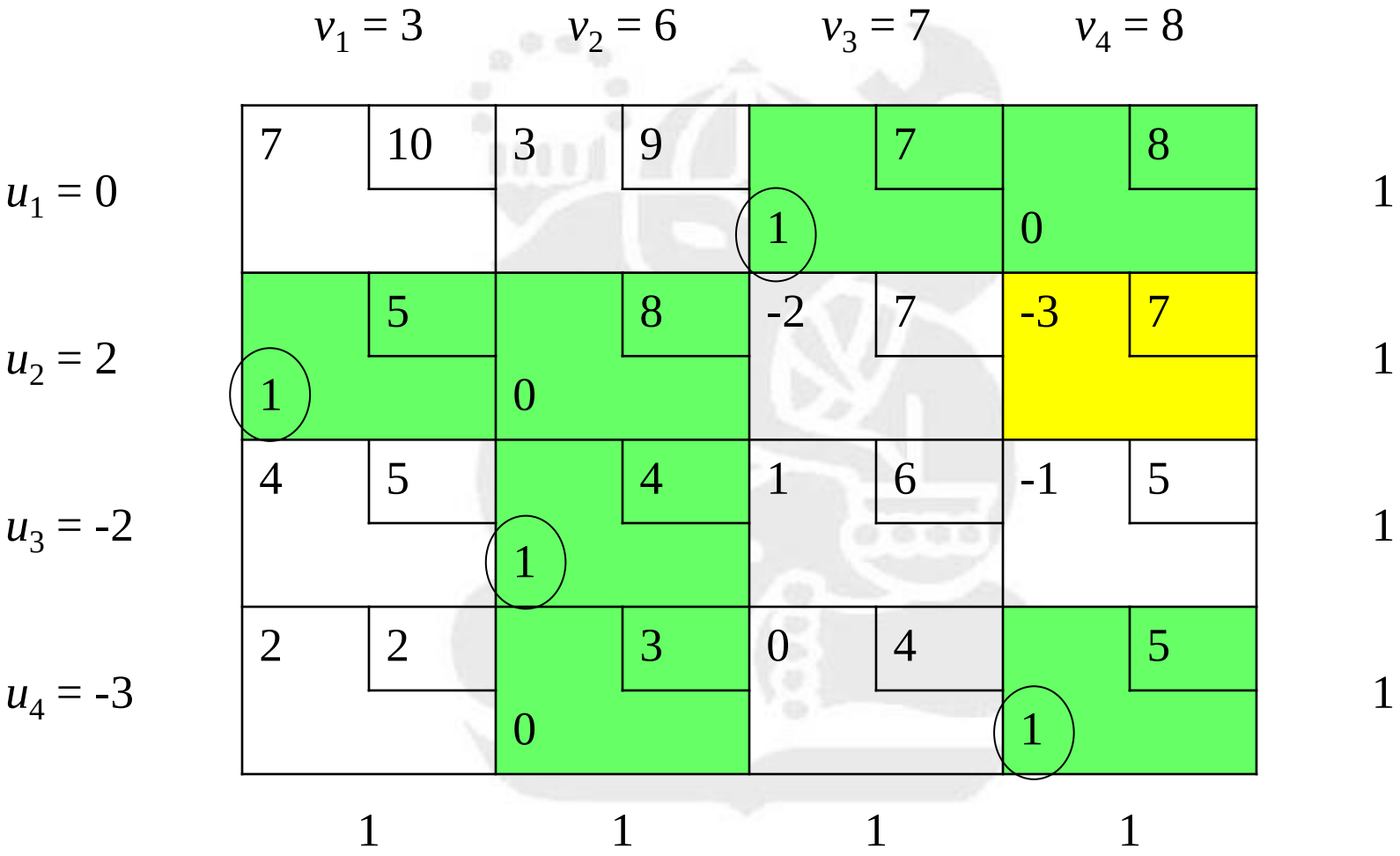


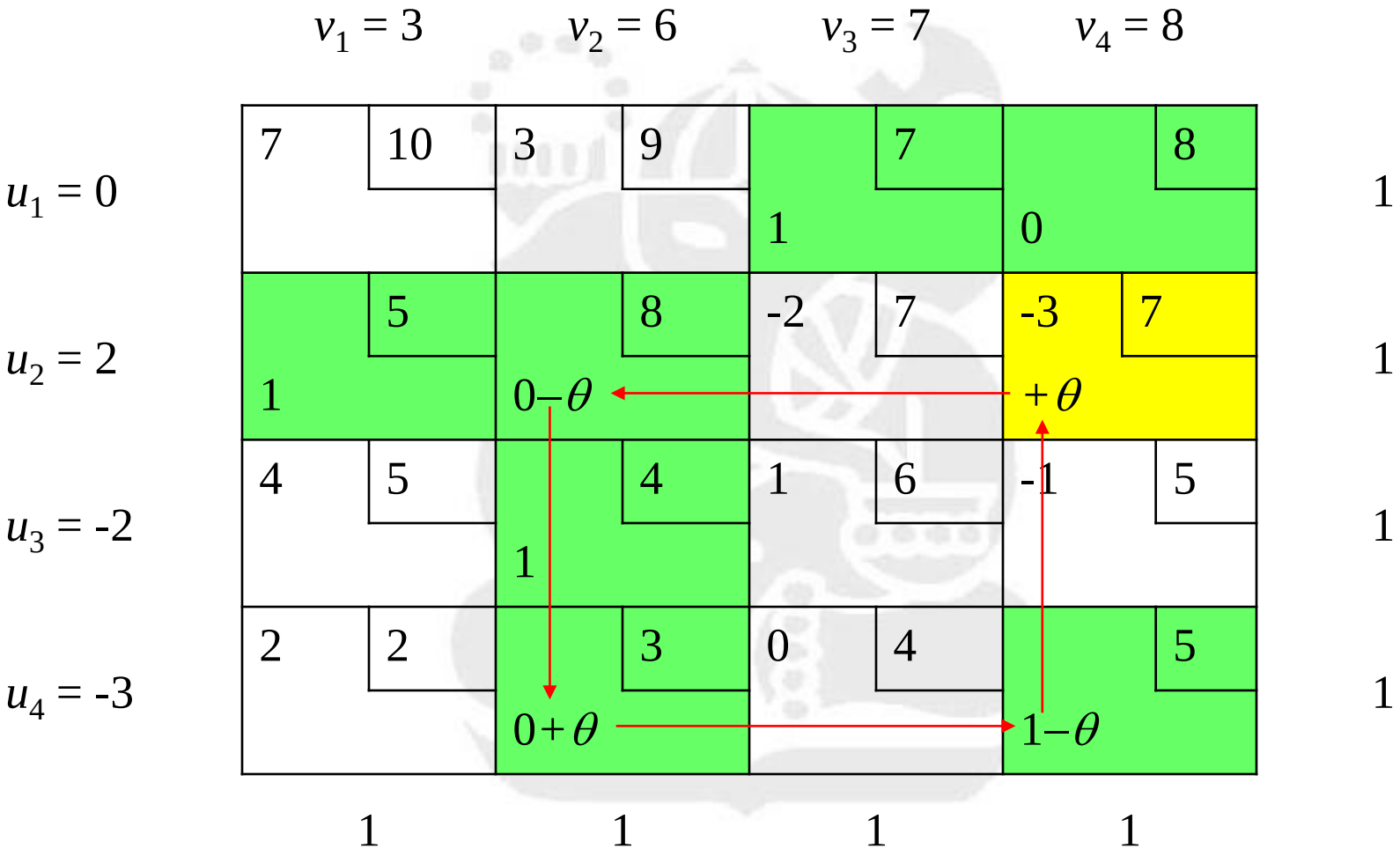

$v_1 = 10$
$v_2 = 13$
$v_3 = 7$
$v_4 = 15$
$u_1 = 0$
$u_2 = -5$
$u_3 = -9$
$u_4 = -10$
10
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-7
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$0-\theta$
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Solusi basis layak baru

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_4$

$Z = 7(1) + 5(1) + 4(1) + 5(1)$
$= 21$

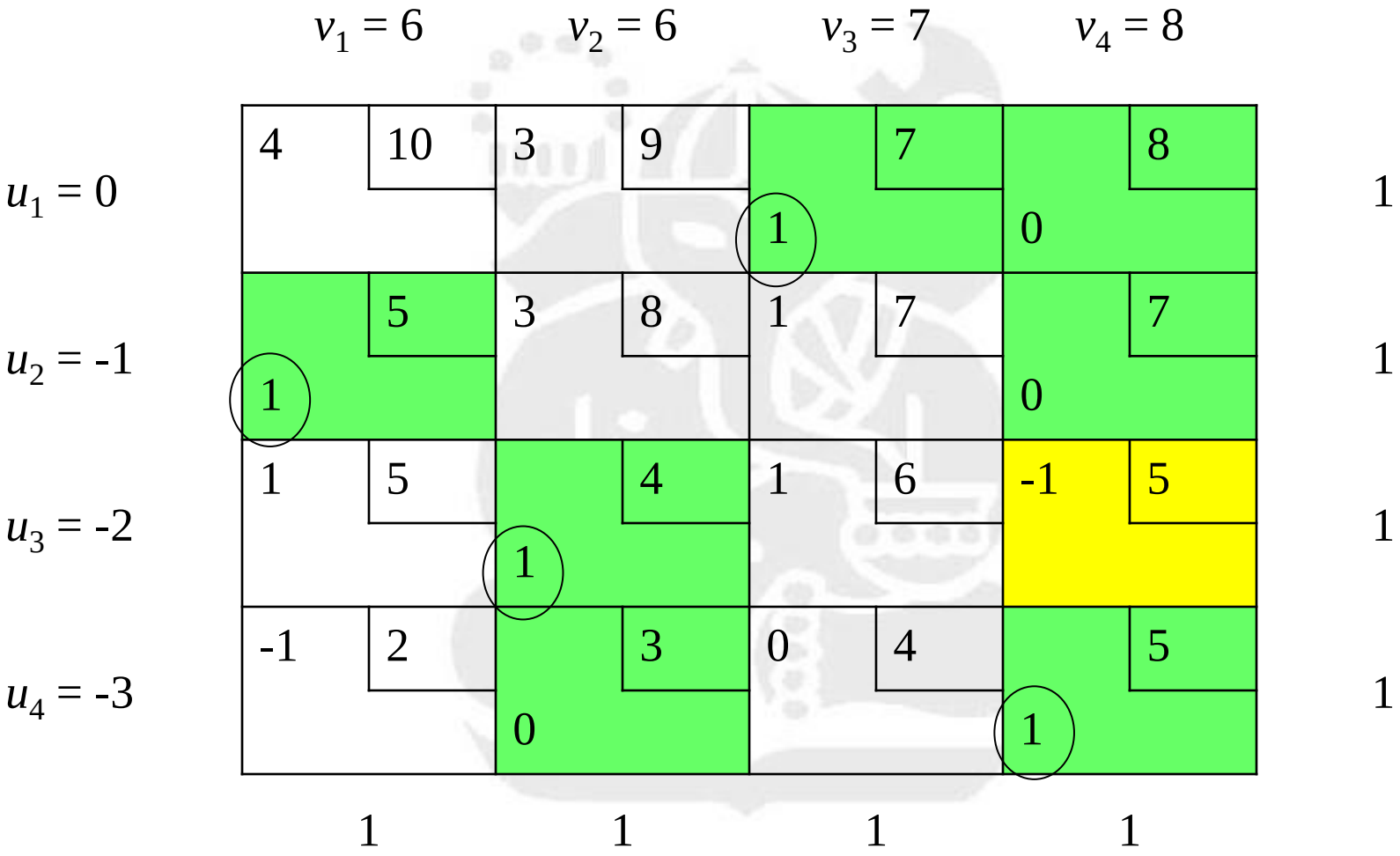


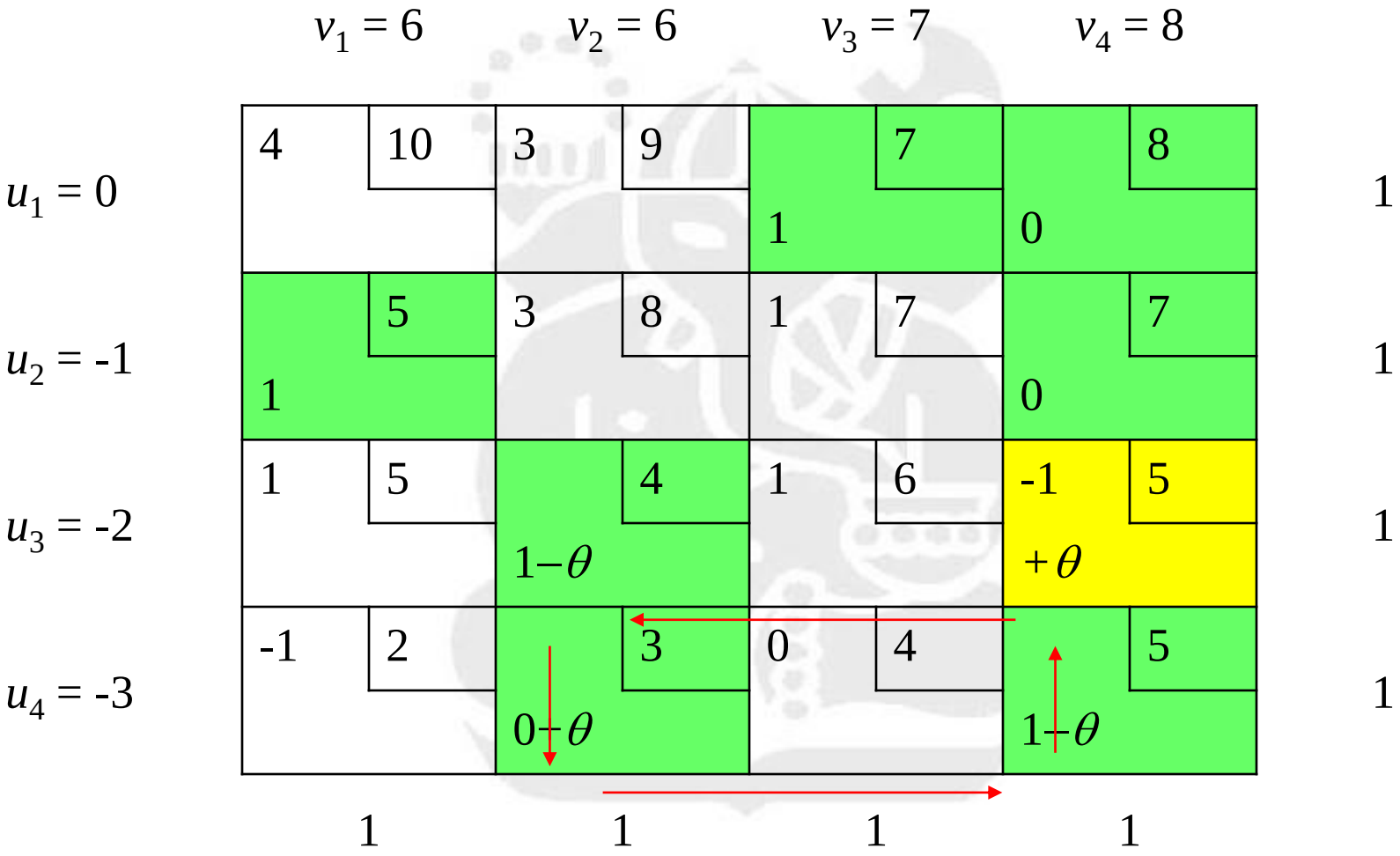

$v_1 = 3$
$v_2 = 6$
$v_3 = 7$
$v_4 = 8$
$u_1 = 0$
$u_2 = 2$
$u_3 = -2$
$u_4 = -3$
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8
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-3
7
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$0-\theta$
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$0+\theta$
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Solusi basis layak baru

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_4$

$Z = 7(1) + 5(1) + 4(1) + 5(1)$
$= 21$

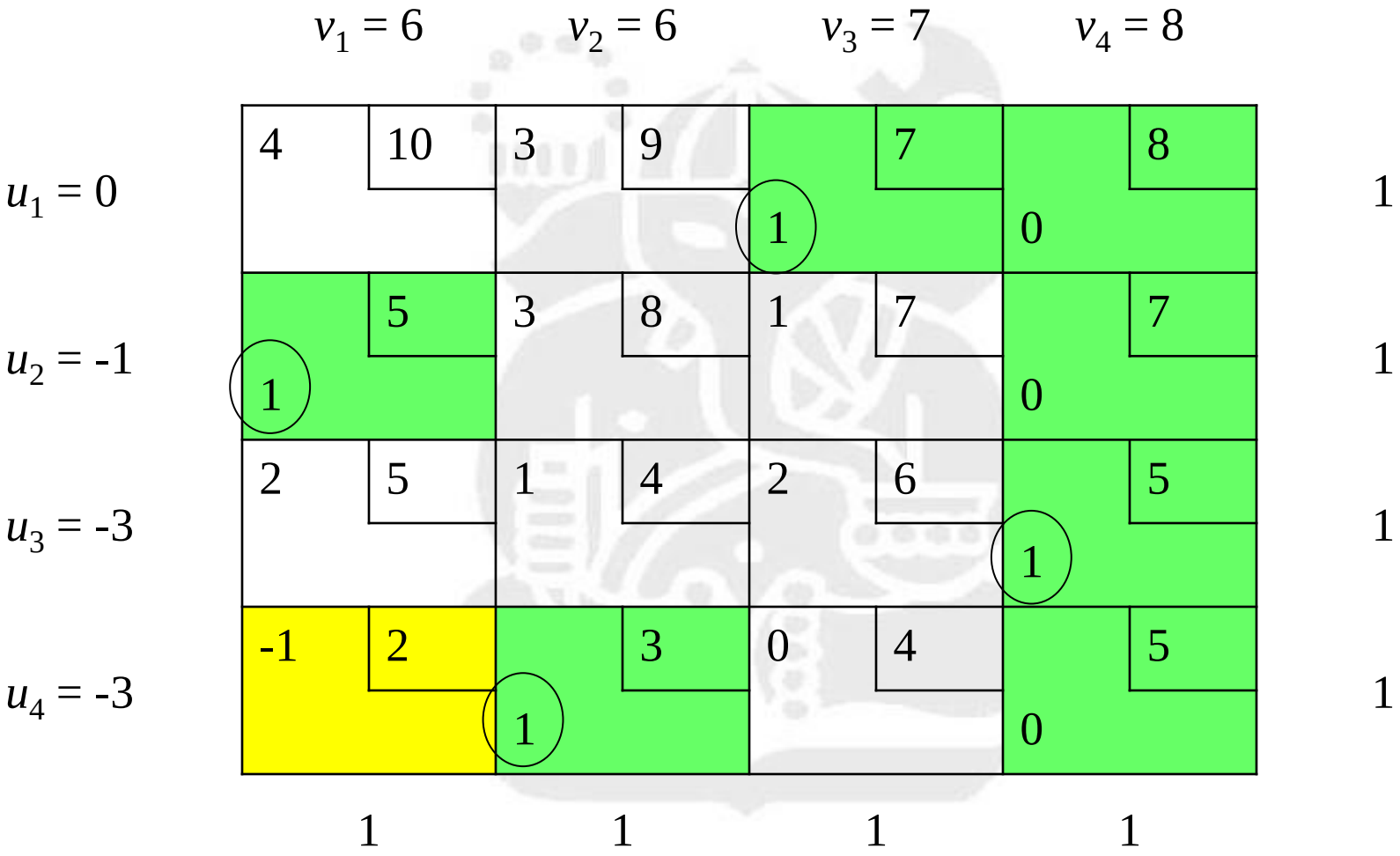


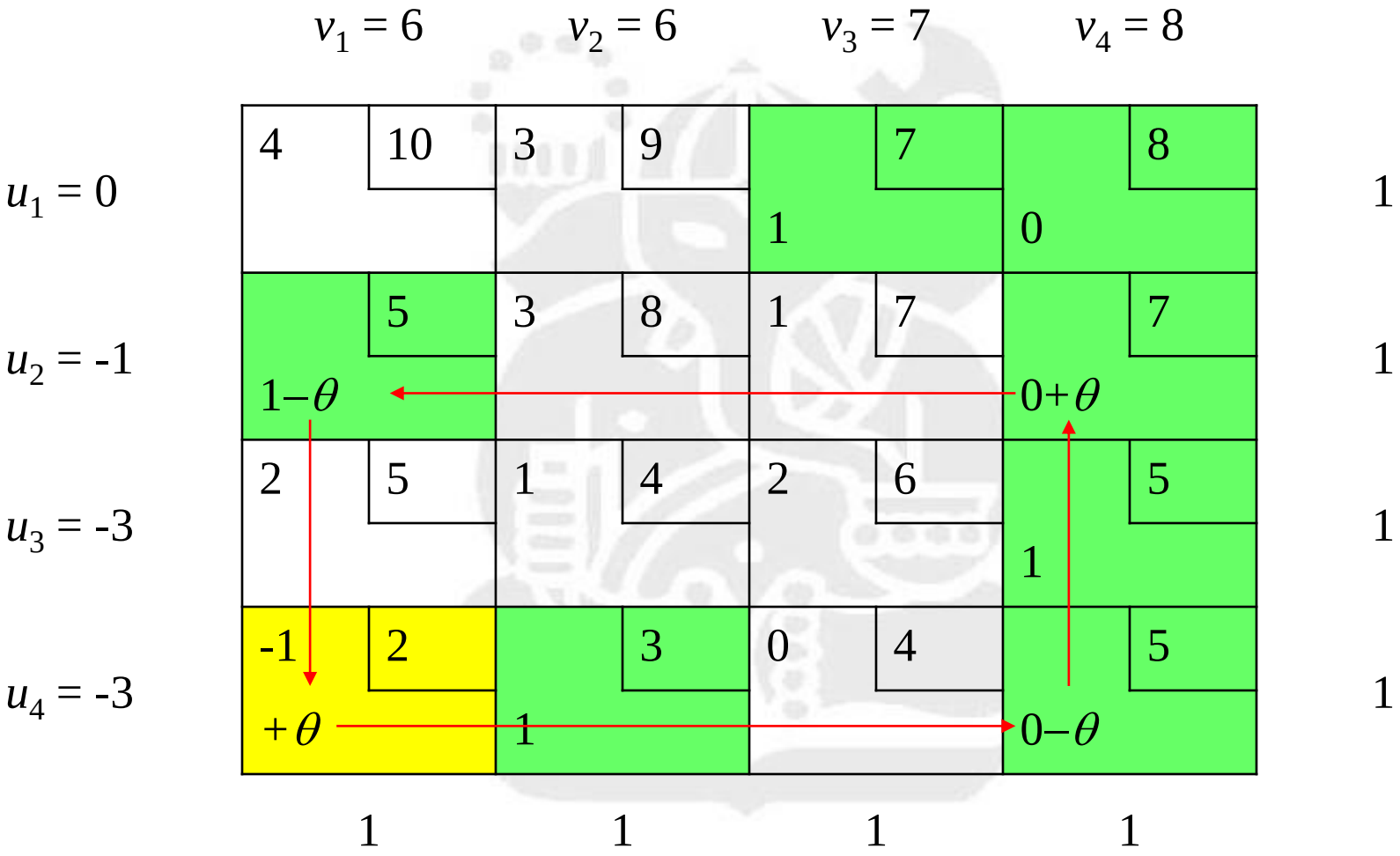

$v_1 = 6$
$v_2 = 6$
$v_3 = 7$
$v_4 = 8$
$u_1 = 0$
$u_2 = -1$
$u_3 = -2$
$u_4 = -3$
4 10 3 9 7 8
1 0 1
5 3 8 1 7 7
1 0 1
1 5 4 1 6 -1 5
$1-\theta$ $+\theta$ 1
-1 2 3 0 4 5
$0+\theta$ $1-\theta$ 1
1 1 1 1

Solusi basis layak baru

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_4$
$M_4 \rightarrow J_2$

$Z = 7(1) + 5(1) + 5(1) + 3(1)$
$= 20$

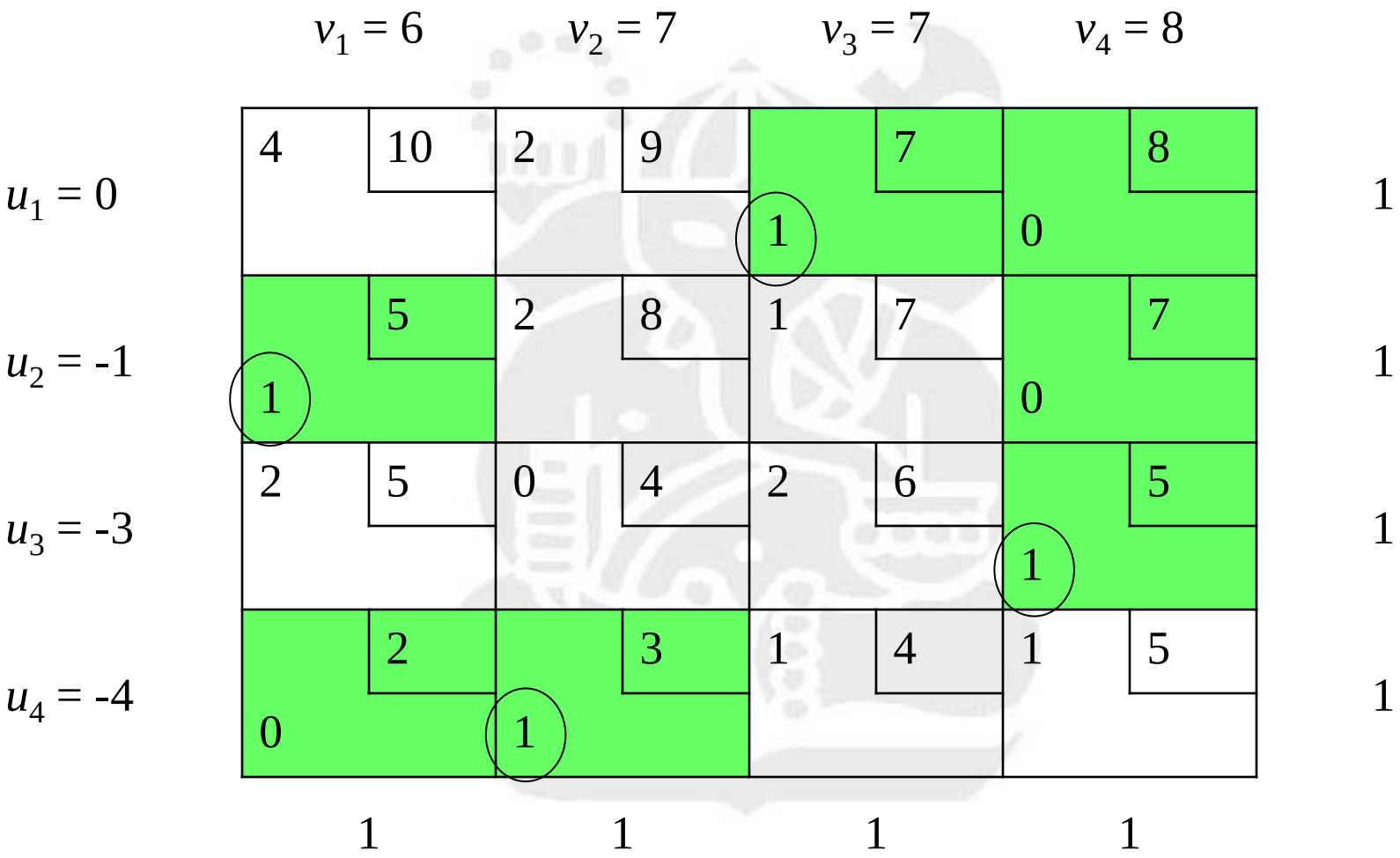


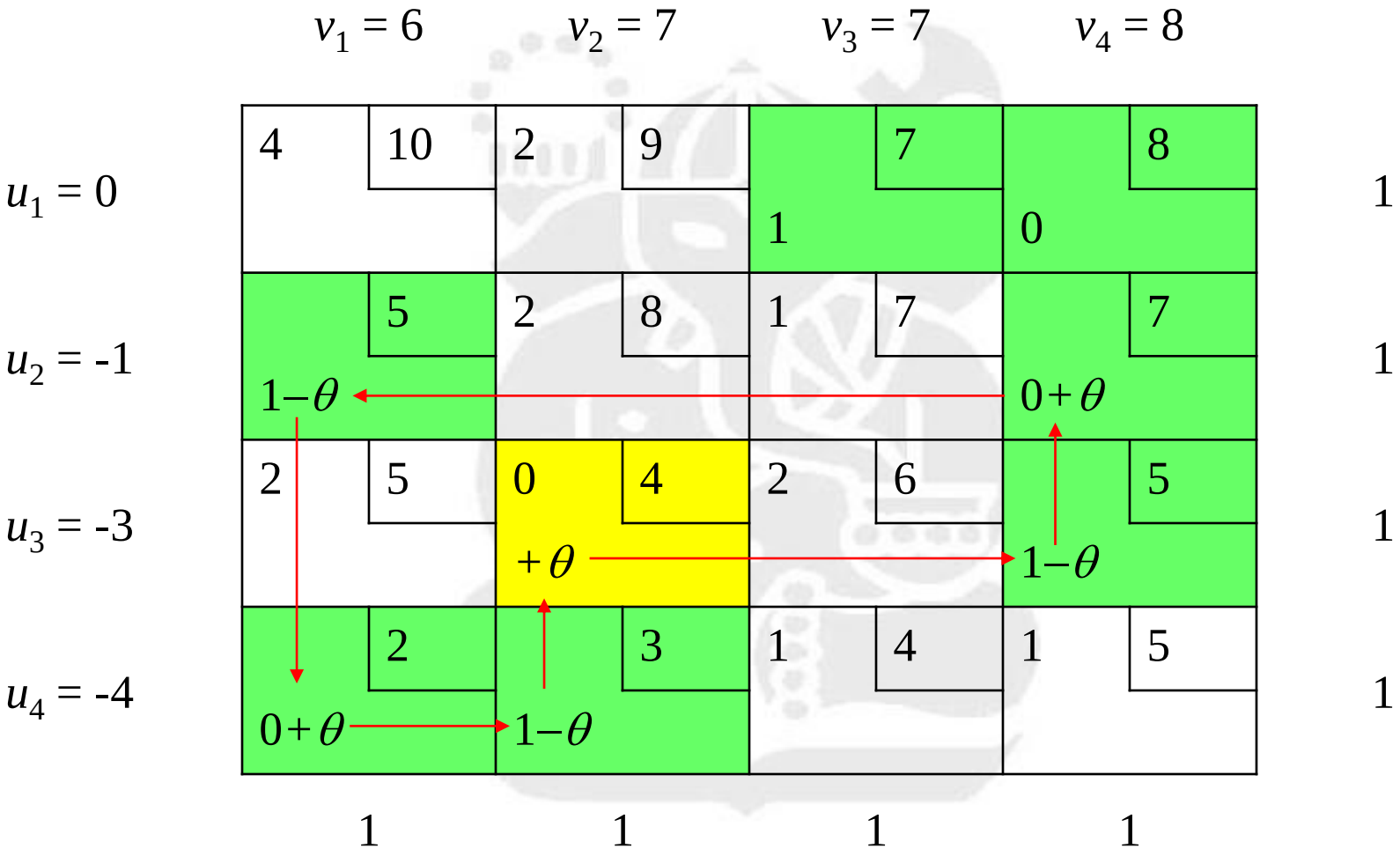

$v_1 = 6$
$v_2 = 6$
$v_3 = 7$
$v_4 = 8$
$u_1 = 0$
$u_2 = -1$
$u_3 = -3$
$u_4 = -3$
4
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$0+\theta$
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1
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2
6
5
1
-1
2
3
0
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$+\theta$
1
$0-\theta$
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Solusi optimal

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_4$
$M_4 \rightarrow J_2$

$Z = 7(1) + 5(1) + 5(1) + 3(1)$
$= 20$

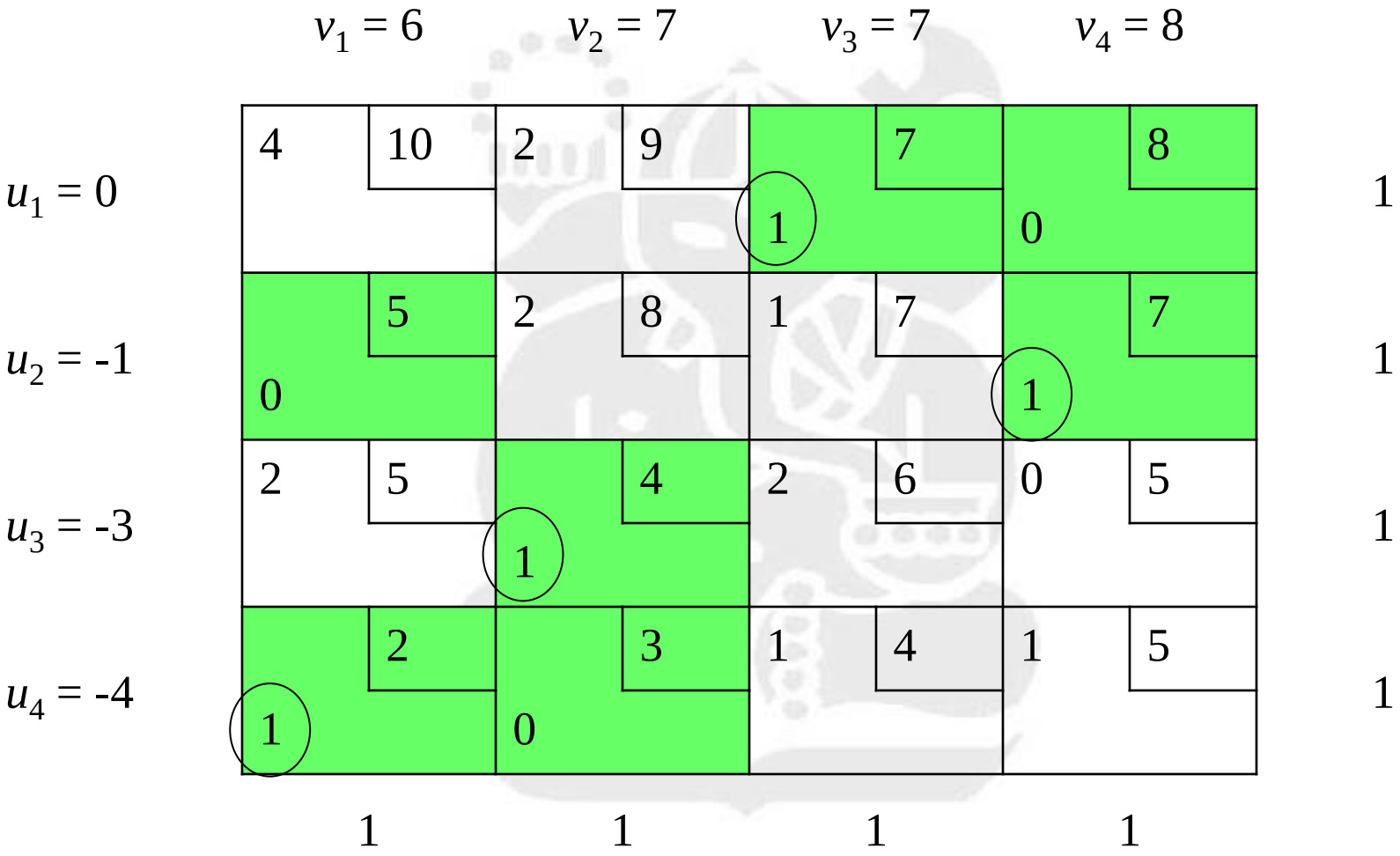



$v_1 = 6$
$v_2 = 7$
$v_3 = 7$
$v_4 = 8$
$u_1 = 0$
$u_2 = -1$
$u_3 = -3$
$u_4 = -4$
4 10
2 9
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2 8
1 7
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$0+\theta$
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2 5
0 4
2 6
5
$+\theta$
$1-\theta$
1
2
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1 4
1 5
$0+\theta$
$1-\theta$
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1 1 1 1

Solusi optimal (Alternatif)

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_4$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_1$

$$Z = 7(1) + 7(1) + 4(1) + 2(1) = 20$$



Teknik *Branch-and-Bound*

Penugasan empat job dan empat mesin

Biaya penugasan :

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

Simpul 1:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	(5)
M_4	(2)	(3)	(4)	5

Total biaya = 14

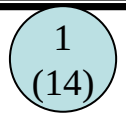
Tidak layak

Batas bawah = 14

Batas atas = ∞

Batas bawah = 17

Batas atas = ∞



Simpul 2:

	J_1	J_2	J_3	J_4
M_1	(10)	9	7	8
M_2	5	8	7	7
M_3	5	4	6	(5)
M_4	2	(3)	(4)	5

Total biaya = 22

Tidak layak

Simpul 3:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	(5)	8	7	7
M_3	5	4	6	(5)
M_4	2	(3)	(4)	5

Total biaya = 17

Tidak layak

Simpul 4:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	(5)	4	6	5
M_4	2	(3)	(4)	(5)

Total biaya = 17

Tidak layak

Simpul 5:

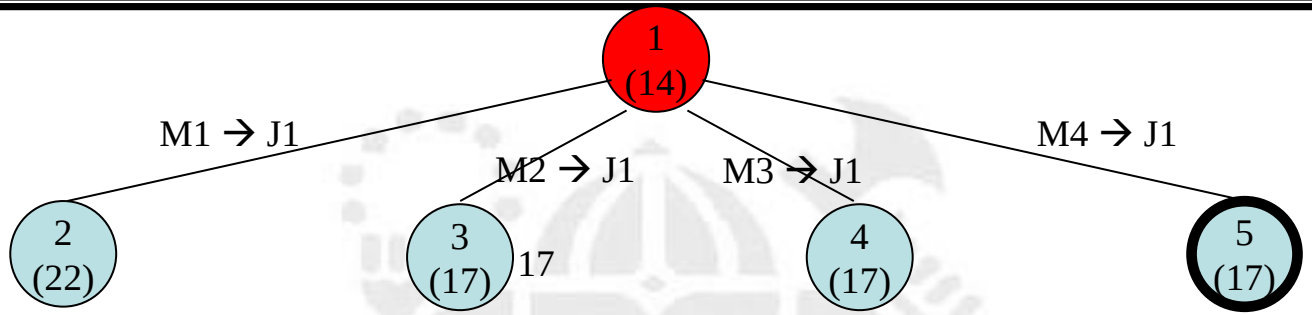
	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	(4)	(6)	(5)
M_4	(2)	3	4	5

Total biaya = 17

Tidak layak

Batas bawah = 17

Batas atas = ∞



Simpul 6:

	J_1	J_2	J_3	J_4
M_1	10	(9)	7	8
M_2	5	8	7	7
M_3	5	4	(6)	(5)
M_4	(2)	3	4	5

Total biaya = 22

Tidak layak

Simpul 7:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	(8)	7	7
M_3	5	4	(6)	(5)
M_4	(2)	3	4	5

Total biaya = 21

Tidak layak

Simpul 8:

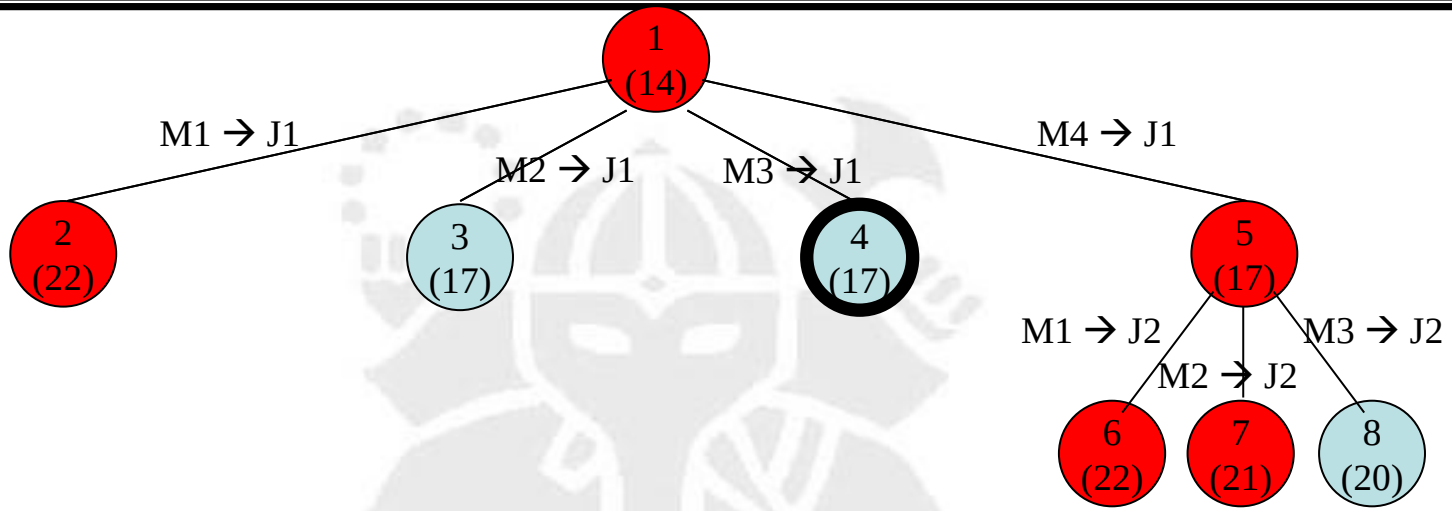
	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	5	8	7	(7)
M_3	5	(4)	6	5
M_4	(2)	3	4	5

Total biaya = 20

Layak

Batas bawah = 17

Batas atas = 20



Simpul 9:

	J_1	J_2	J_3	J_4
M_1	10	(9)	7	8
M_2	5	8	7	7
M_3	(5)	4	6	5
M_4	2	3	(4)	(5)

Total biaya = 23

Tidak Layak

Simpul 10:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	(8)	7	7
M_3	(5)	4	6	5
M_4	2	3	(4)	(5)

Total biaya = 22

Tidak Layak

Simpul 11:

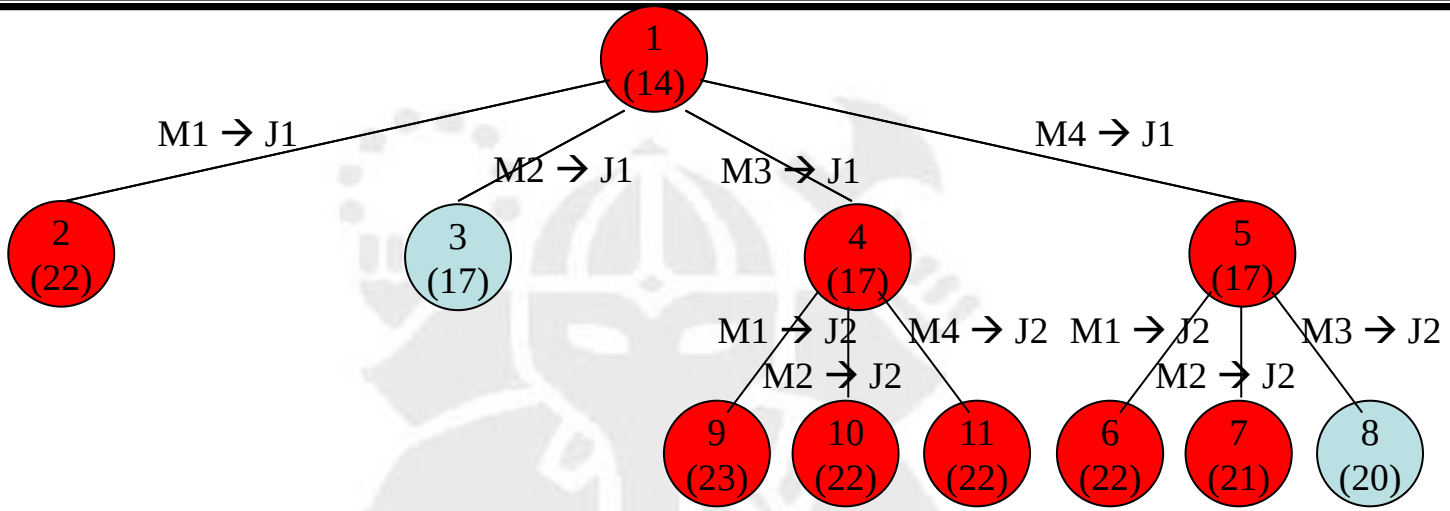
	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	5	8	7	(7)
M_3	(5)	4	6	5
M_4	2	(3)	4	5

Total biaya = 22

Layak

Batas bawah = 17

Batas atas = 20



Simpul 12:

	J_1	J_2	J_3	J_4
M_1	10	(9)	7	8
M_2	(5)	8	7	7
M_3	5	4	6	(5)
M_4	2	3	(4)	5

Total biaya = 23

Layak

Simpul 13:

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	(5)	8	7	7
M_3	5	(4)	6	5
M_4	2	3	(4)	(5)

Total biaya = 18

Tidak layak

Simpul 14:

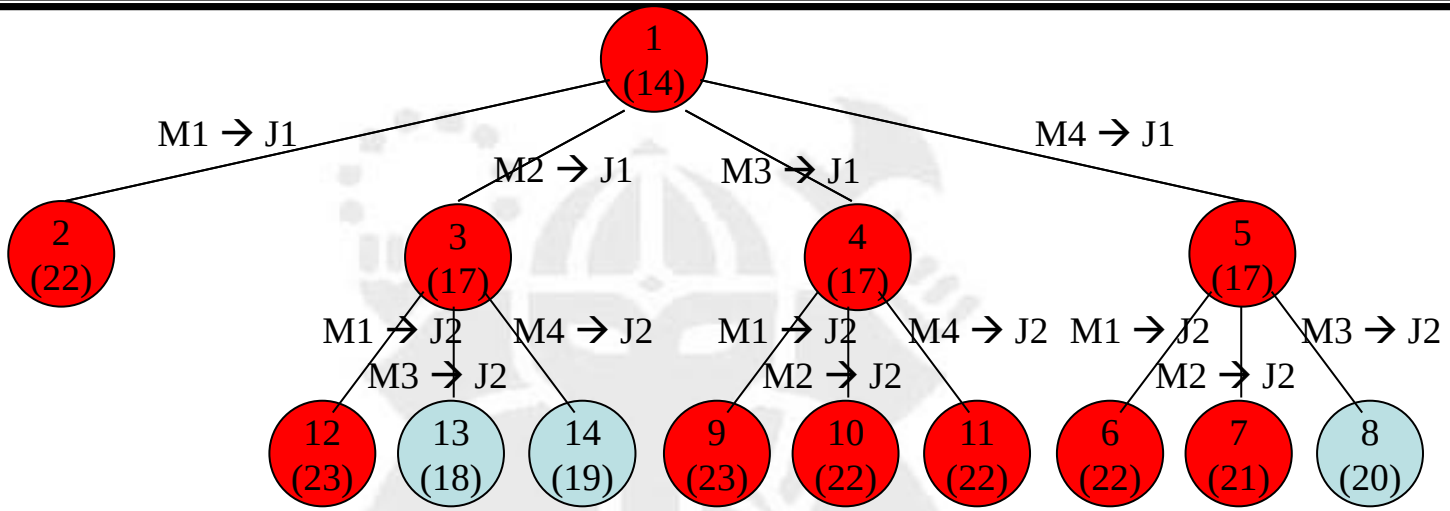
	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	(5)	8	7	7
M_3	5	4	(6)	(5)
M_4	2	(3)	4	5

Total biaya = 19

Tidak layak

Batas bawah = 18

Batas atas = 20



Simpul 15:

	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	(5)	8	7	7
M_3	5	(4)	6	5
M_4	2	3	4	(5)

Total biaya = 21

Layak

Simpul 16:

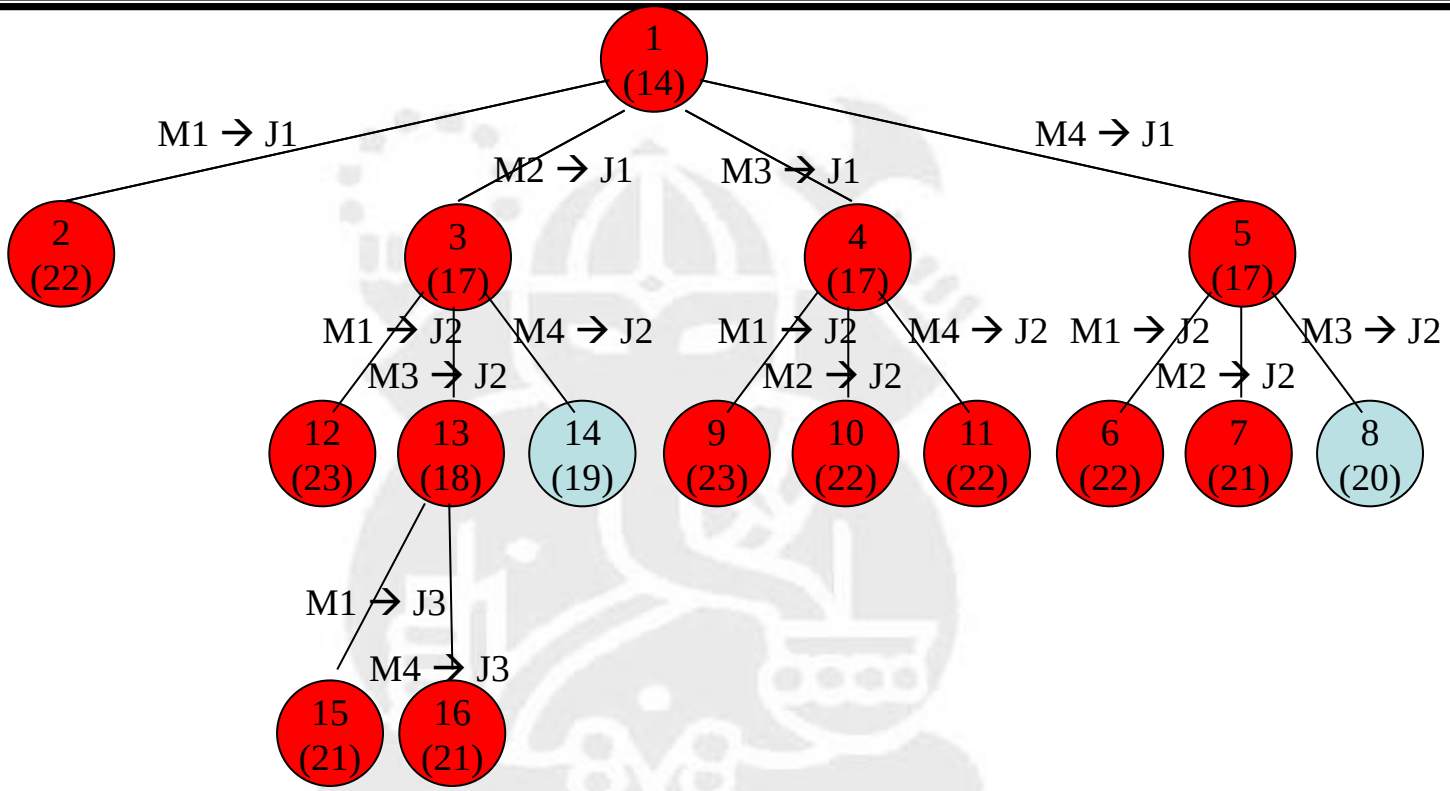
	J_1	J_2	J_3	J_4
M_1	10	9	7	(8)
M_2	(5)	8	7	7
M_3	5	(4)	6	5
M_4	2	3	(4)	5

Total biaya = 21

Layak

Batas bawah = 19

Batas atas = 20



Simpul 17:

	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	(5)	8	7	7
M_3	5	4	6	(5)
M_4	2	(3)	4	5

Total biaya = 20

Layak

Langkah 18:

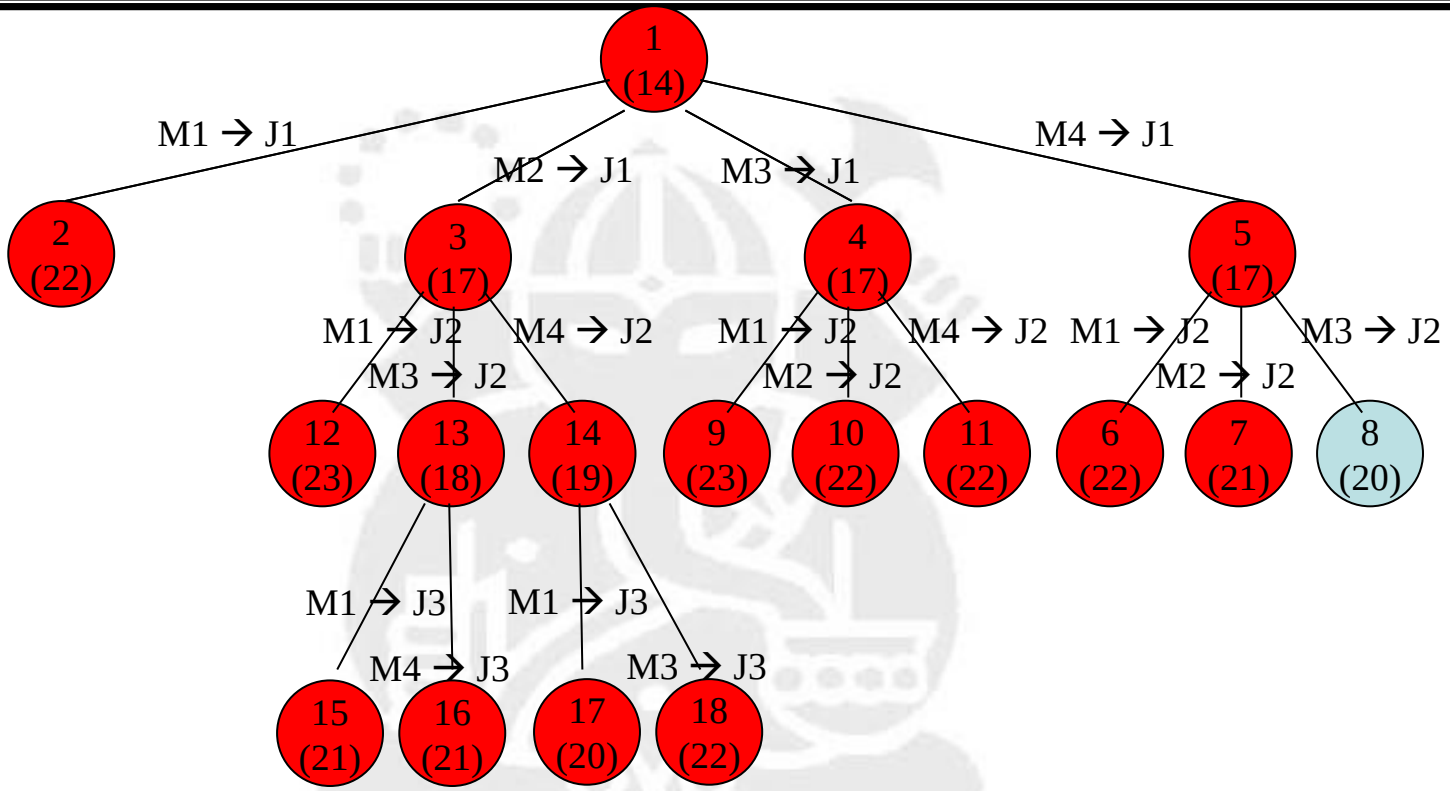
	J_1	J_2	J_3	J_4
M_1	10	9	7	(8)
M_2	(5)	8	7	7
M_3	5	4	(6)	5
M_4	2	(3)	4	5

Total biaya = 22

Layak

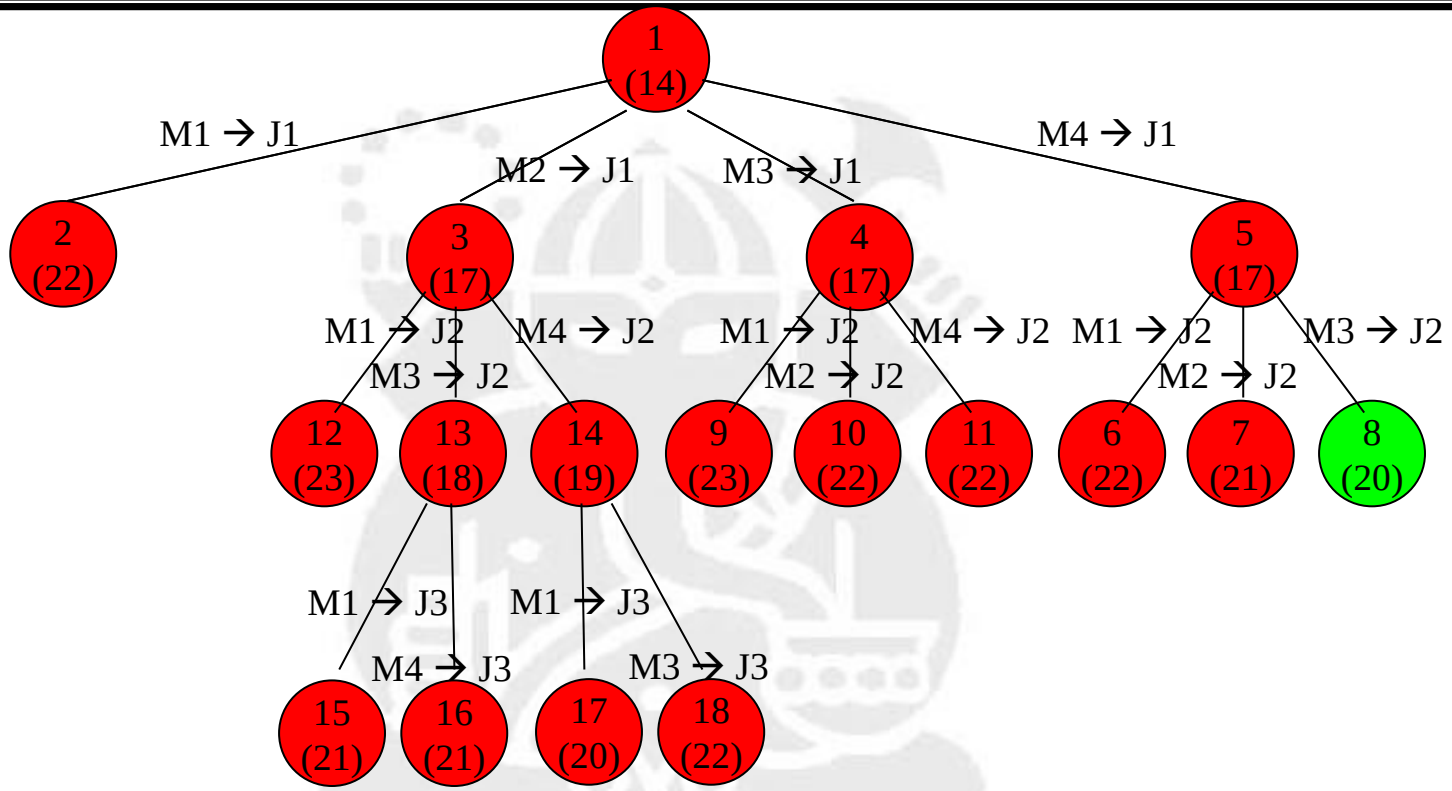
Batas bawah = 20

Batas atas = 20



Batas bawah = 20

Batas atas = 20



Solusi Optimal

Solusi Optimal

Simpul 8:

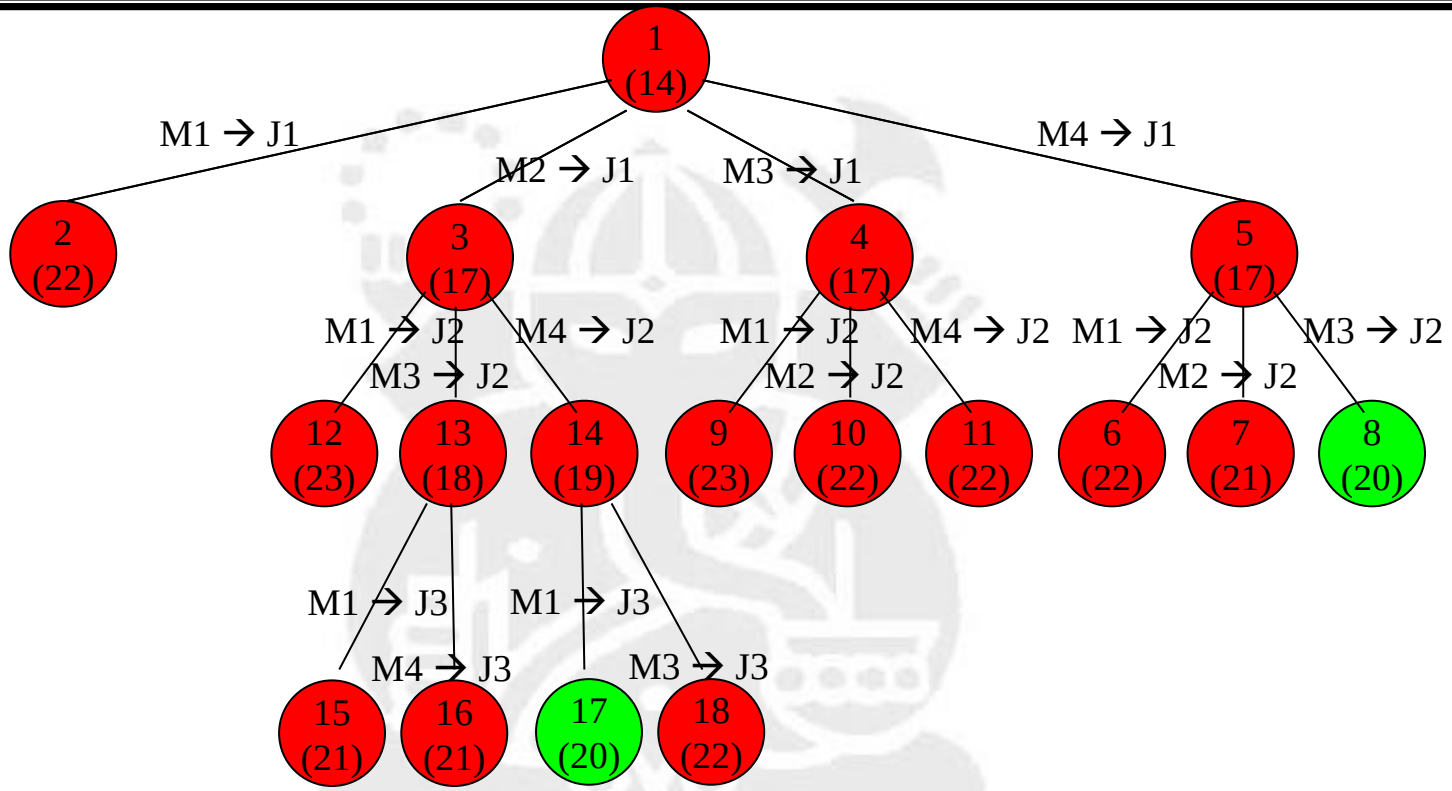
	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	5	8	7	(7)
M_3	5	(4)	6	5
M_4	(2)	3	4	5

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_4$
$M_3 \rightarrow J_2$
$M_4 \rightarrow J_1$

Biaya penugasan total $Z = 7 + 7 + 4 + 2 = 20$

Batas bawah = 20

Batas atas = 20



Solusi Optimal

Solusi Optimal (alternatif)

Simpul 17:

	J_1	J_2	J_3	J_4
M_1	10	9	(7)	8
M_2	(5)	8	7	7
M_3	5	4	6	(5)
M_4	2	(3)	4	5

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_4$
$M_4 \rightarrow J_2$

Biaya penugasan total $Z = 7 + 5 + 5 + 3 = 20$

Metode Hungarian (1)

- Asumsi adalah semua elemen biaya (c_{ij}) tak negatif.
- Prinsip dasar:
  - Solusi optimal penugasan tidak terpengaruh jika suatu konstanta ditambahkan atau dikurangkan dari sebarang baris atau kolom dari matriks biaya.

Metode Hungarian (2)

$$Z = \sum_i \sum_j c_{ij} x_{ij} = \sum_i \sum_j \left(c_{ij} - p_i - q_j\right) x_{ij}$$

$$= \sum_i \sum_j c_{ij} x_{ij} - \sum_i \sum_j p_i x_{ij} - \sum_i \sum_j q_j x_{ij}$$

$$= \sum_i \sum_j c_{ij} x_{ij} - \sum_i \sum_j p_i x_{ij} - \sum_i \sum_j q_j x_{ij}$$

$$= \sum_i \sum_j c_{ij} x_{ij} - \sum_i p_i \left(\sum_j x_{ij} \right) - \sum_i q_j \left(\sum_i x_{ij} \right)$$

$$= \sum_i \sum_j c_{ij} x_{ij} - \sum_i p_i (1) - \sum_i q_j (1)$$

$$= \sum_i \sum_j c_{ij} x_{ij} - \text{konstanta}$$

Struktur Algoritma dari Metode Hungarian

- Langkah 0:
  - Matriks biaya
- Langkah 1:
  - Reduksi baris. Jika diperoleh solusi layak maka berhenti dan diperoleh solusi optimal. Jika sebaliknya, lanjutkan ke Langkah 2.
- Langkah 2:
  - Reduksi kolom. Jika diperoleh solusi layak maka berhenti dan diperoleh solusi optimal. Jika sebaliknya, lanjutkan ke Langkah 3.
- Langkah 3
  - Modifikasi matriks biaya. Algoritma berhenti apabila telah diperoleh solusi layak (dan optimal).

Contoh #1

Penugasan empat job dan empat mesin

Biaya penugasan :

	J_1	J_2	J_3	J_4
M_1	10	9	8	7
M_2	3	4	5	6
M_3	2	1	1	2
M_4	4	3	3	6

Contoh # 1
Reduksi Baris

	J_1	J_2	J_3	J_4
M_1	3	2	1	(0)
M_2	(0)	1	2	3
M_3	1	0	(0)	1
M_4	1	(0)	2	3

Solusi Layak

Contoh # 1
Solusi Optimal

	J_1	J_2	J_3	J_4
M_1	3	2	1	(0)
M_2	(0)	1	2	3
M_3	1	0	(0)	1
M_4	1	(0)	2	3

$M_1 \rightarrow J_4$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_3$
$M_4 \rightarrow J_2$

Biaya penugasan total $Z = 7 + 3 + 1 + 3 = 14$

Contoh #2

Penugasan empat job dan empat mesin

Biaya penugasan :

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

Contoh #2

	J_1	J_2	J_3	J_4
M_1	10	9	8	7
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

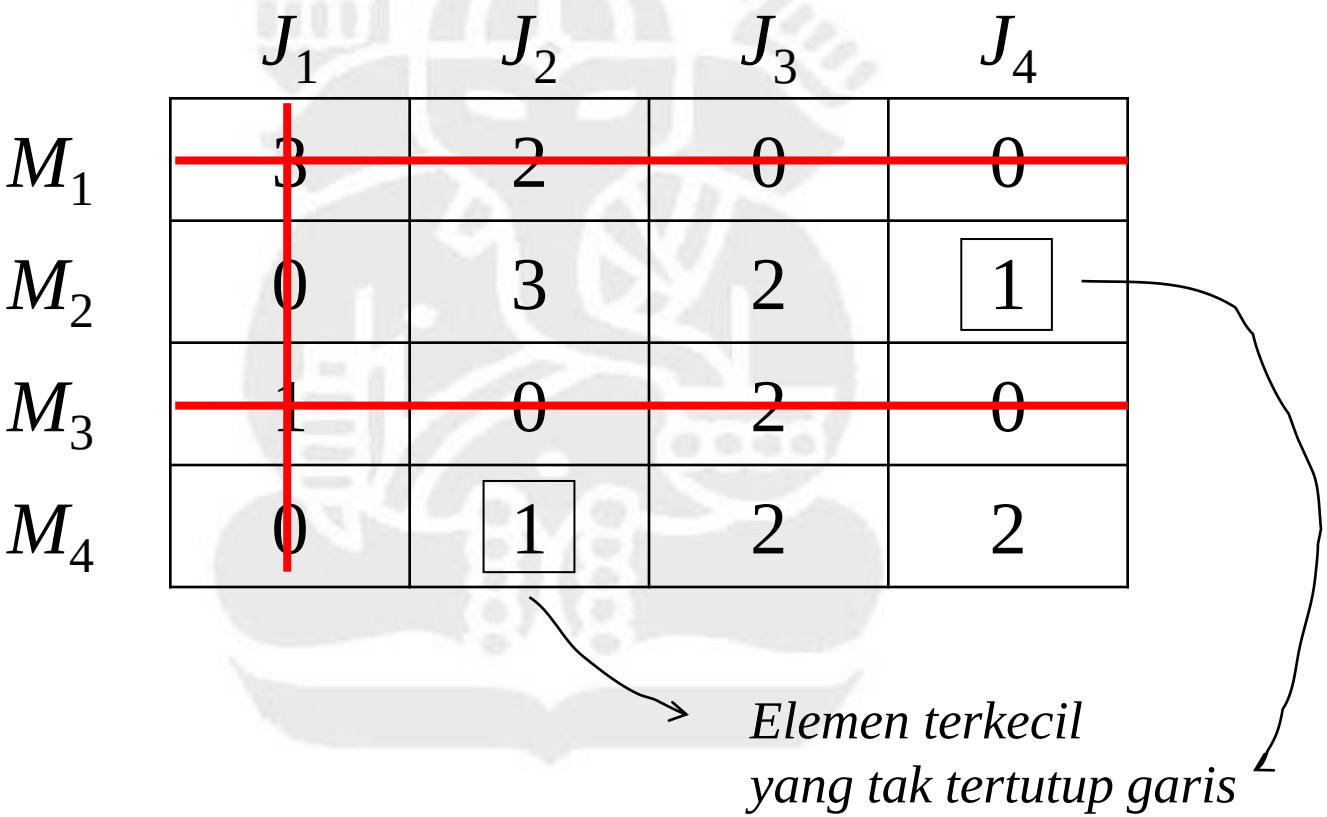
Contoh # 2
Reduksi Baris

	J_1	J_2	J_3	J_4
M_1	3	2	0	1
M_2	0	3	2	2
M_3	1	0	2	1
M_4	0	1	2	3

Contoh #2
Reduksi Kolom

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	0	1	2	2

Contoh #2
Modifikasi Matrix



Contoh #2
Solusi Layak

	J_1	J_2	J_3	J_4
M_1	4	2	0	0
M_2	0	2	1	0
M_3	1	0	2	0
M_4	0	0	1	1

Contoh #2
Solusi Layak (dan Optimal)

	J_1	J_2	J_3	J_4
M_1	4	2	(0)	0
M_2	(0)	2	1	0
M_3	1	0	2	(0)
M_4	0	(0)	1	1

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_4$
$M_4 \rightarrow J_2$

Biaya penugasan total $Z = 7 + 5 + 5 + 3 = 20$

Contoh #2
Solusi Optimal Alternatif

	J_1	J_2	J_3	J_4
M_1	4	2	(0)	0
M_2	0	2	1	(0)
M_3	1	(0)	2	0
M_4	(0)	0	1	1

$M_1 \rightarrow J_3$
$M_2 \rightarrow J_1$
$M_3 \rightarrow J_4$
$M_4 \rightarrow J_2$

Biaya penugasan total $Z = 7 + 7 + 4 + 2 = 20$

Masalah Maksimisasi

- Langkah 1
  - Konversikan ke masalah minimasi dengan mengalikan semua elemen (c_{ij}) dari matrix biaya penugasan dengan -1.
- Langkah 2
  - Jika beberapa elemen matrix biaya adalah negatif maka tambahan suatu bilangan positif yang besar untuk masing-masing baris dan kolom sehingga elemen-elemen biaya menjadi positif.
- Langkah 3
  - Terapkan metode Hungarian.